

BL54L10 and BL54L15 Series

Datasheet

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Introduction

Overview

This document describes key hardware aspects of the BL54L15 and BL54L10. This document is intended to assist device manufacturers and related parties with the integration of this radio into their host devices. Data in this document is drawn from several sources. For full documentation on the BL54L15 and BL54L10, visit:

<https://www.ezurio.com/bl54l15>

<https://www.ezurio.com/bl54l10-series>

General Description

Experience a new pinnacle of performance, efficiency, and security with our new BL54L15 and BL54L10 series, built on Nordic Semiconductor's powerful **nRF54L QFN** silicon. Elevating what you know and love from the nRF52 series, this next generation redefines Bluetooth LE and 802.15.4 solutions. Unleashing enhanced processing power, expanded memory, and innovative peripherals, the BL54L15 is the ultimate choice for low power connectivity.

Powered by **Nordic's nRF54L15 and nRF54L10** SoC, our compact BL54L15 and BL54L10 modules deliver secure and robust Bluetooth LE and 802.15.4 with flexible programming via Nordic's nRF Connect SDK or Ezurio Canvas Software Suite.

Featuring a **128MHz ARM Cortex M33** and **128MHz RISC-V coprocessor**, supported by 1.5 MB non-volatile memory and 256 KB RAM for BL54L15, 1.0 MB non-volatile memory and 192 KB RAM for BL54L10, the BL54L15 and BL54L10 modules offer double the processing power (vs prior BL654 – nRF52840). The BL54L15 / L10 series brings out all nRF54L15/ L10 hardware features and capabilities including up to **+7 dBm** transmit power, **1.7V – 3.6V** supply considerations, and **NFC A-Tag** implementation.

It's further enhanced with state-of-the-art security and is designed for PSA Certified level 3 and supports services such as Secure Boot, Secure Firmware Update, Secure Storage plus protection from physical attacks.

Note: BL54L15/ BL54L10 hardware provides all functionality of the nRF54L15/ nRF54L10 chipset used in the module design. This is a hardware datasheet only – it does not cover the software aspects of the BL54L15 and BL54L10. This is to acknowledge that information in this datasheet is referenced from the nRF54L15/ nRF54L10 datasheet.

This datasheet is subject to change. Please [contact Ezurio](#) for further information.

Application Areas

- Building Automation
- Security
- Medical Peripherals
- Industrial Sensors

Features & Benefits

The BL54L15 device features and benefits are described below.

- Nordic nRF54L15/ nRF54L10 – 6x6 mm QFN48 with 31 GPIOs utilized.
- Multi-protocol support: Bluetooth LE, 802.15.4 (Thread & Matter)
- Cortex M33 processor core: 128 MHz ARM Cortex M33
- RISC-V co-processor core: 128 MHz VPR
- Memory:
 - 1.5MB non-volatile memory, 256 KB RAM for BL54L15
 - 1.0MB non-volatile memory, 192 KB RAM for BL54L10
- High Speed Peripherals: - HS-SPI/UART, software defined peripherals on 128 MHz VPR, GPIO - 1x 64 MHz Port, 1.7 – 3.6V, 11 GPIOs
- Low Leakage Peripherals: 2x QDEC, 7x Timer, Global RTC, 2x WDT, NFC A-Tag, TEMP, I2S, COMP, 3x PWM, LPCOMP, 14-bit 8CH ADC, 5x TWI/SPI/UART, GPIO (2x 16 MHz Port (P0, P1), 20 GPIO's, 1.7-3.6V, 31 GPIO)
- Antenna choice – integrated pre-certified PCB Trace antenna or external antenna support via MHF4 connector
- Ultra-small footprint (14 mm x 10 mm x 1.6 mm)
- Extended Industrial Temperature Rating (-40° to +105 °C)

Specification Summary

Processor / SoC / Chipset

MPU	Application Core (High Performance): • Arm Cortex-M33 with DSP, FPU, TrustZone support. • 1524KB non-volatile RRAM (BL54L15) • 1022KB non-volatile RRAM (BL54L10) • 256KB RAM (BL54L15) • 192KB RAM (BL54L10) • L1 cache • 128MHz clock • Uses voltage and clock frequency scaling
VPU	Software-Defined Peripheral Core (Ultra-Low Power): • RISC-V CPU (VPR) fast lightweight peripheral processor (FLPR) dedicated for software defined peripherals • 16MHz clock
Wireless	Nordic nRF54L15 SoC

Bluetooth

Standards	Bluetooth Core 6.2
Bluetooth Radio	Bluetooth Low Energy • Bluetooth Low Energy Coded PHY (Long range) • Bluetooth Constant Tone Extension (Direction-finding) • IEEE 802.15.4
Bluetooth LE Features	• Channel Sounding • GATT client & GATT server – Any adopted/custom services • Central/Peripheral roles • Mesh networking • LE CODED • LE 2M • LE Secure Connections • Extended Advertising • LE Power Control • DTM Firmware (Test Modes)

PHY	2405–2480 MHz IEEE 802.15.4-2006 radio transceiver, implementing IEEE 802.15.4-2006 compliant • 250kbps, 2450MHz, O-QPSK PHY • Channels 11-26. Channel 11 2405MHz and CH26 2480MHz. • Clear channel assessment (CCA) • Energy detection (ED) scan • CRC generation
Proprietary Radio Modes	2402–2480 MHz Nordic proprietary 1Mbps and 2Mbps modes radio transceiver • 1Mbps nRF proprietary mode (ideal transmitter) • 2Mbps nRF proprietary mode (ideal transmitter) • 4Mbps nRF proprietary mode (ideal transmitter)
Max Transmit Power	+7 dBm - Conducted 453-00001, 453-00225 (Integrated antenna) (Exclude antenna gain) +7 dBm - Conducted 453-00044, 453-00226 (External antenna) (Test at MHF4 connector)
Min Transmit Power	-40 dBm
Receive Sensitivity(1) (≤ 37 byte packet for BLE)	• BLE 1 Mbps (BER=1E-3): -94 dBm typical • BLE 2 Mbps: -92 dBm typical • BLE 125 kbps: -102 dBm typical • BLE 500 kbps: -98 dBm typical • IEEE 802.15.4-2020 250kbps: TBD dBm typical
Supported Data Rates	• 2.402 - 2.480 GHz for BLE (CH0 to CH39) • 2.405 - 2.480 GHz for IEEE 802.15.4-2006 PHY (CH11 to CH26) 1 Mbps BLE (over-the-air) • 2 Mbps BLE (over-the-air) • 125 kbps BLE (over-the-air) • 500 kbps BLE (over-the-air) • 250 kbps IEEE 802.15.4-2020 (over-the-air) • Nordic proprietary 1Mbps, 2Mbps and 4Mbps modes (over-the-air)
Link Budget	101 dB @ BLE 1 Mbps (TX power used is 7dBm) 109 dB @ BLE 125 kbps (TX power used is 7dBm)

NFC

NFC-Type A Listen mode compliant	Based on NFC forum specification: 13.56 MHz, Data rate 106 kbps, NFC Type2 and Type 4 emulation Modes of Operation: Disable, Sense, Activated Use Cases: Touch-to-Pair with NFC, NFC enabled Out-of-Band Pairing
Security	Designed for PSA Certified Level 3 with Secure Boot, Secure Firmware Update, and Secure Storage. Integrated tamper sensors detect attacks and take action, and cryptographic accelerators are hardened against side-channel attacks.

Radio Performance

Antenna Options	Internal: - PCB Trace antenna – on-board (453-00001 variant for BL54L15, 453-00225 variant for BL54L10) External: - Connection via on module IPEX MHF4 (453-00044 variant for BL54L15, 453-00226 variant for BL54L10)
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Interfaces

Memory Interfaces	- Non-volatile memory (RRAM): 1524 KB - Random access memory (RAM): 256 KB
Peripheral Interface	31x Multifunction I/O lines
HS-UART	Application Core (High Performance): 1x

GPIO	Up to 31 multifunction GPIO's: - Wakeup Pins: 20x - Analog Input Pins: 10x Application Core (High Performance): 64MHz 1.7-3.6V GPIO port - P2.00-P2.10 Software defined peripheral Core (Ultra-Low Power): 16MHz 1.7-3.6V GPIO port - P1.00-P1.14; - P0.00-P0.04
ADC (14-bit)	Application Core (High Performance): 14-bit 31.25KS/s with oversampling 12-bit 250KS/s 10-bit 2MS/s - AIN0-AIN7 pins upto 8 programmable gain channels
Global RTC (GRTC)	Software defined peripheral Core (Ultra-Low Power): - Implements full real time clock and calendar as shared system time. Can run in System OFF mode. - Ultra low power, 1us resolution, 52bits wide, uses 16MHz clock, 32.76kHz when other power modes.
RTC	Software defined peripheral Core (Ultra-Low Power): 2x low power runs off LFCLK
PWM	Software defined peripheral Core (Ultra-Low Power): 3x 4channel PWM
I2C	2x
SPI/UART/TWI	Software defined peripheral Core (Ultra-Low Power): 4x

TEMP	Software defined peripheral Core (Ultra-Low Power): • 1x Temperature sensor • Temperature range equal to operating temperature range
I2S	Software defined peripheral Core (Ultra-Low Power): • 1x I2S (Inter-IC sound interface)
PDM	Software defined peripheral Core (Ultra-Low Power): • 1x PDM (Pulse code modulation interface) for digital microphones
TIMER	Software defined peripheral Core (Ultra-Low Power): • 7x Timer (32bit)
QDEC	Software defined peripheral Core (Ultra-Low Power): • 2x QDEC (Quadrature decoder)
COMP	Software defined peripheral Core (Ultra-Low Power): • 1x COMP (comparator)
LPCOMP	Software defined peripheral Core (Ultra-Low Power): 1x LPCOMP (low power comparator)
WDT	Software defined peripheral Core (Ultra-Low Power): • 2x WDT (Watchdog timer)
NFC A-Tag	Software defined peripheral Core (Ultra-Low Power): • 1x

Wakeup Pins	Software defined peripheral Core (Ultra-Low Power): • 20x
Ext. 32.768 KHz Crystal	Software defined peripheral Core (Ultra-Low Power): • Not needed for normal radio operation. • Optionally, connect +/-20ppm accuracy crystal for more accurate protocol timing. Fit associated load capacitor for crystal or use nRF54L15/nRF54L10 internal load capacitor, which is configurable as 4 pF to 18 pF in 0.5 pF steps on pins XL1, XL2.

Power

Supply Voltage	Normal Voltage Mode (VDD_nRF): 1.7V-3.6V (Internal DCDC convertor or LDO)
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Mechanical

Dimensions	14mm x 10mm x 1.6mm Pad Pitch – 0.75 mm Pad Type – Three rows of pads (LGA - Land Grid Array).
Weight	<1 gram

Software

Security	Designed for PSA Certified Level 3 with Secure Boot, Secure Firmware Update, and Secure Storage. Integrated tamper sensors detect attacks and take action, and cryptographic accelerators are hardened against side-channel attacks. • Active tamper shield pin pairs (in/out) x4 • Glitch detectors • Key management unit • Encryption accelerator
Programmability Options	Via SWD (JTAG) 2 wire interface: • Nordic nRFConnect SDK: Software/Support available from Nordic directly https://devzone.nordicsemi.com/ • Canvas SW Suite: Software/ Support available from https://www.ezurio.com/canvas/software-suite
Firmware Update	Via SWD (JTAG) 2 wire interface or UART

Environmental

Operating Temperature	-40 °C to +105 °C
Storage Temperature	-40 °C to +105 °C
Lead Free	Lead-free and RoHS Compliant

Certifications

Regulatory Compliance	FCC/ISED/CE/MIC/RCM/UKCA/KC
Bluetooth SIG	Full Bluetooth SIG Declaration ID

Development

Development Kit	Development kit per module SKU (453-00001-K1 and 453-00044-K1 respectively, BL54L15 only) Utilise the above development kits for projects planned with BL54L10.
Development Tools	- Nordic nRFConnect - Android and iOS applications - UART firmware upgrade - Xbit Tools and utilities
Debug	ITM parallel trace

Warranty

Warranty Terms	One Year Warranty
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Functional Descriptions

Programming & Firmware

The BL54L15/BL54L10 module is shipped from Ezurio manufacturing facilities with no firmware programmed.

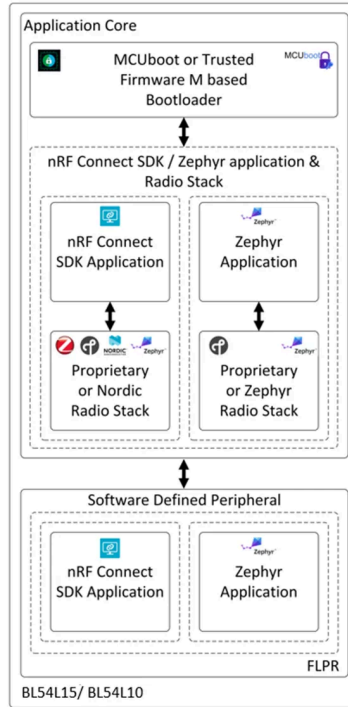
BL54L15 BL54L10 Firmware Options

Firmware for use with the BL54L15/BL54L10 can be divided into the following types.

- Bootloader** – This is the application that resides on the Application Core used to perform firmware updates of the Application and FLPR cores.

The MCU Boot or Trusted Firmware M Bootloaders are recommended to be used as the basis for the BL54L15 Bootloader functionality.

- **Application** – This is the main application running on the Application core. It interfaces with the integrated radio stack(s) and provides supplementary functionality in addition to the time critical activities performed for radio activity.
- **Software Defined Peripheral** – This is the application running on the RISC-V FLPR core.



Clocks

HFXO - 32MHz crystal oscillator and nRF54L15/nRF54L10 internal load capacitor mandatory setting

The BL54L15/BL54L10 module contains the 32 MHz crystal, but the load capacitors to create 32MHz crystal oscillator circuit are inside the nRF54L15/nRF54L10 chipset. Customer can set the internal nRF54L15/nRF54L10 capacitors from 4 pF to 17 pF in 0.25pF if needed or use default value without changing.

The 32 MHz crystal inside the BL54L15/BL54L10 module is a high accuracy crystal (± 15 ppm at room temperature) that helps with radio operation and reducing power consumption in the active modes.

LFCLK – Low Frequency clock source

There are four possibilities (see figure 3) for the low frequency clock (LFCLK) and options are:

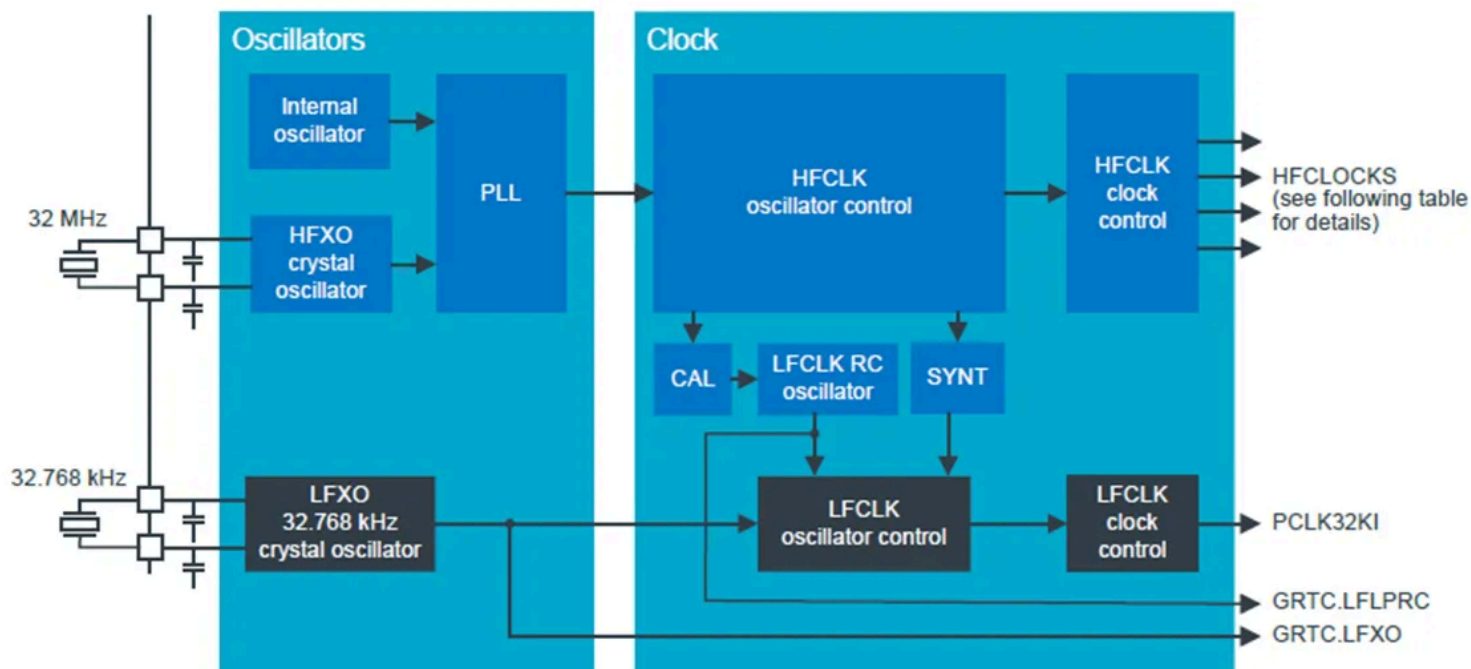
LFRC (32.768kHz RC oscillator): The Internal 32.768 kHz RC oscillator (LFRC) is fully embedded in nRF54L15/nRF54L10 (and does not require additional external components) with an accuracy ± 250 ppm (after calibration of LFRC at least every eight seconds using the HFXO as a reference oscillator).

LF XO (32.768kHz crystal oscillator): For higher LFCLK accuracy (greater than ± 250 ppm accuracy is required), the low frequency crystal oscillator (LF XO) must be used. To use LF XO, a 32.768kHz crystal must be connected between the XL1 and XL2 pins and the load capacitance between each crystal terminal and ground. Optionally internal (to nRF54L15/nRF54L10) capacitor of maximum 18 pF in 0.5 pF steps are provided on pins XL1 and XL2.

Low frequency (32.768 kHz) external source: The 32.768 kHz oscillator (LFXO) is designed to work with external sources

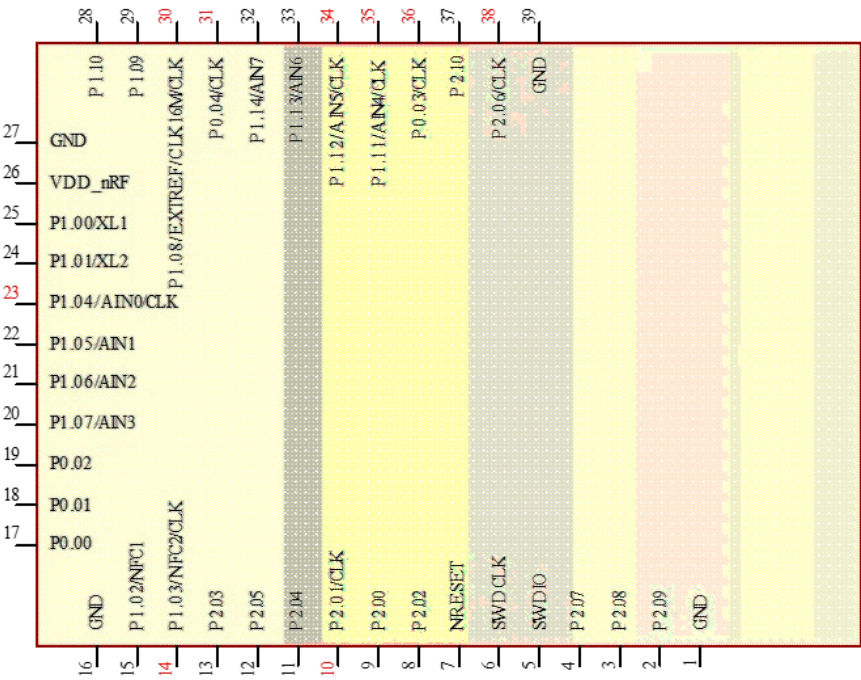
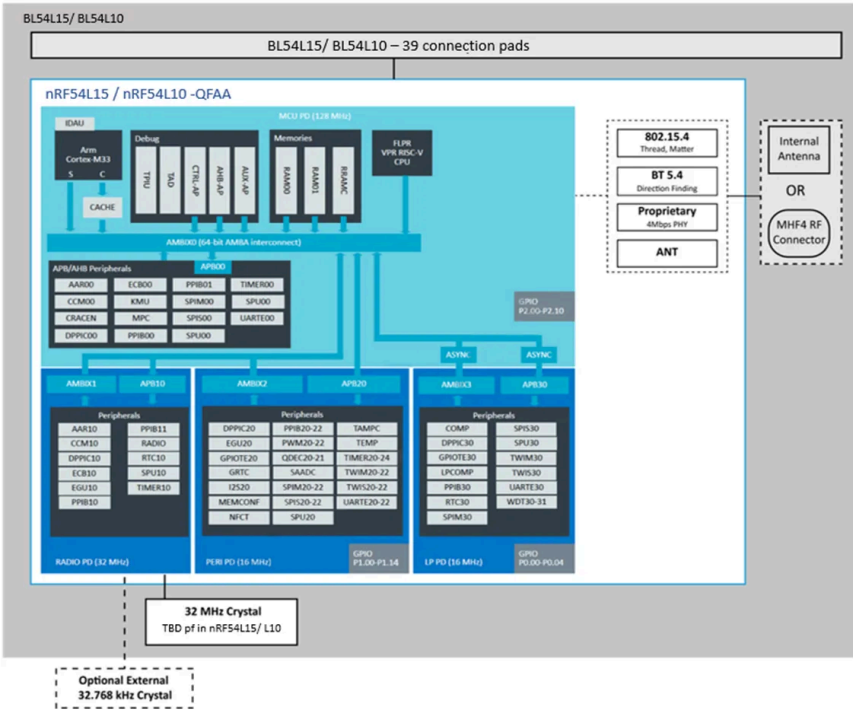
LFSYNTH (32.768kHz Synthesised clock) from HFCLK (LFSYNTH): The LFCLK can be synthesised from the HFCLK source. LFSYNTH depends on the HFCLK to run. The accuracy of the LFCLK clock with LFSYNTH as a source assumes the accuracy of the HFCLK. If high accuracy is required, the HFCLK must be generated from the HFXO. Using the LFSYNTH clock removes the requirement for an external 32.768kHz crystal but increases the average power consumption as the HFCLK will be turned on in the system.

Other Internal Clocks



Hardware Architecture

Block Diagrams



Pin-Out

Pin #	Pin Name (bold coloured pins for clock for interfaces, trace)	nRF54L15 nRF54L10 QFN48 Pin	nRF54L15/ nRF54L10 QFN48 Name	Description	Example usage
1	GND	-	VSS		
2	P2.09	20	P2.09/TRACEDA TA[2]/SPI.SDI/S DO	General purpose I/O Trace data SPIM SDI SPIS SDI UARTE CTS	Trace SPIM00/SPIM21 SPIS00/SPIS21 UARTE00/UART E21
3	P2.08	19	P2.08/TRACEDA TA[1]/SPI.SDI/S DO	General purpose I/O Trace data SPIM SDO SPIS SDO UARTE TXD	Trace SPIM00/SPIM21 SPIS00/SPIS21 UARTE00/UART E21
4	P2.07	18	P2.07/TRACEDA TA[0]/SWO/SPI. DCX	General purpose I/O Trace data Serial wire output (SWO) SPIM DCX UARTE RXD	Trace Trace SPIM00/SPIM21 UARTE00/UART E21
5	SWDIO	25	SWDIO	Serial Wire Debug IO for debug and programming	

6	SWDCLK	26	SWDCLK	Serial Wire Debug clock input for debug and programming	
7	NRESET	30	nRESET	Pin RESET with internal pull-up resistor (13k Ohms). System Reset (Active Low).	
8	P2.02	13	P2.02/SPI.SDI/SDO	General purpose I/O SPIM SDO SPIS SDO UARTE TXD QSPI D0 Serial wire output (SWO)	SPIM00/SPIM20 SPIS00/SPIS20 UARTE00/UARTE20 FLPR Trace
9	P2.00	11	P2.00/SPI.DCX	General purpose I/O SPIM DCX UARTE RXD QSPI D3	SPIM00/SPIM20 UARTE00/UARTE20 FLPR (QSPI)
10	P2.01/CLK	12	P2.01/SPI.SCK	General purpose I/O SPIM SCK SPIS SCK QSPI SCK	Clock pin SPIM00/SPIM20 SPIS00/SPIS20 FLPR

11	P2.04	15	P2.04/SPI.SDI/S DO	General purpose I/O SPIM SDI SPIS SDI UARTE CTS QSPI D1	SPIM00/SPIM20 SPIS00/SPIS20 UARTE00/UART E20 FLPR
12	P2.05	16	P2.05/SPI.CS	General purpose I/O SPIM CS UARTE RTS QSPI CS	SPIM00/SPIM20 UARTE00/UART E20 FLPR
13	P2.03	14	P2.03	General purpose I/O QSPI D2	FLPR
14	P1.03/NFC2/CL K	4	P1.03/NFC2	General purpose I/O Dedicated pin for NFC input	Clock pin
15	P1.02/NFC1	3	P1.02/NFC1	General purpose I/O Dedicated pin for NFC input	
16	GND				
17	P0.00	23	P0.00	General purpose I/O	
18	P0.01	24	P0.01	General purpose I/O	
19	P0.02	27	P0.02	General purpose I/O	

20	P1.07/AIN3	8	P1.07/AIN3/TAMPC	General purpose I/O TAMPC active shield 1 input Analog input	TAMPC
21	P1.06/AIN2	7	P1.06/AIN2/TAMPC	General purpose I/O TAMPC active shield 1 output Analog input	TAMPC
22	P1.05/AIN1	6	P1.05/AIN1/TAMPC	General purpose I/O TAMPC active shield 0 input RADIO DFEGPIO Analog input	TAMPC RADIO
23	P1.04/AIN0/CLK	5	P1.04/AIN0/TAMPC	General purpose I/O TAMPC active shield 0 output Analog input	Clock pin TAMPC
24	P1.01/XL2	2	P1.01/XL2	General purpose I/O	Ezurio Devkit: Optional 32.768kHz crystal pad XL2, XL1 and associated 9pF load capacitor inside nRF54L15 chipset.
25	P1.00/XL1	1	P1.00/XL1	General purpose I/O Connection for 32.768kHz crystal	

General purpose I/O	Ezurio Devkit: Optional 32.768kHz crystal pad XL2, XL1 and associated 9pF load capacitor inside nRF54L15 chipset.
Connection for 32.768kHz crystal	Ezurio Devkit: Optional 32.768kHz crystal pad XL2, XL1 and associated 9pF load capacitor inside nRF54L15 chipset.

26	VDD_nRF	48	VDD	1.7V-3.6V input for Normal Voltage Mode where connect external supply to VDD_nRF (pin26).	
27	GND				
28	P1.10	38	P1.10/TAMPC	General purpose I/O TAMPC active shield 2 input RADIO DFEGPIO	TAMPC RADIO
29	P1.09	37	P1.09/TAMPC	General purpose I/O TAMPC active shield 2 output RADIO DFEGPIOO	TAMPC RADIO
30	P1.08/EXTREF/CLK16M/CLK	9	P1.08/CLK16M/TAMPC	General purpose I/O GRTC HF clock output External reference for SAADC	Clock pin
31	P0.04/CLK	29	P0.04/GRTC_CLKOUT32K	General purpose I/O GRTC LF clock output	Clock pin GRTC

32	P1.14/AIN7	42	P1.14/AIN7	General purpose I/O RADIO DFEGPIO Analog input	RADIO
33	P1.13/AIN6	41	P1.13/AIN6	General purpose I/O RADIO DFEGPIO Analog input	RADIO
34	P1.12/AIN5/CLK	40	P1.12/AIN5	General purpose I/O TAMPC active shield 3 input RADIO DFEGPIO Analog input	Clock pin TAMPC RADIO
35	P1.11/AIN4/CLK	39	P1.11/AIN4	General purpose I/O TAMPC active shield 3 output RADIO DFEGPIO Analog input	Clock pin TAMPC RADIO
36	P0.03/CLK	28	P0.03/GRTC_PWMOUT	General purpose I/O GRTC PWM output	Clock pin GRTC
37	P2.10	21	P2.10/TRACEDA TA[3]/SPIM.CS	General purpose I/O Trace SPIM CS UARTE RTS	Trace SPIM00/SPIM21 UARTE00/UARTE21

38	P2.06/CLK	17	P2.06/TRACECLK/SPI.SCK	General purpose I/O SPIM SCK SPIS SCK Trace clock	Clock pin SPIM00/SPIM21 SPIS00/SPIS21 Trace
39	GND				

Notes:**Note 1 (GPIO):**

- **GPIO = General Input or Output (GPIO level voltage tracks VDD pin). AIN = Analog input.**
- If GPIO is selected as an input, ensure the input is not floating (which can cause current consumption to drive with time in low power modes (such as System ON Idle), by selecting the internal pull up or pull down.
- **Must connect all GND pads to host board PCB GND plane.**

Note2 (Clock for serial interfaces or trace):

- Some peripherals (SPI, TWI, PDM, I2S, TRACE, GRTC) have clock signals. Dedicated clock pins have been optimized to ensure correct timing relationship between clock and data signal for these peripherals. Pins that can be used as clock signals are shown with pin name in red colour.
- The peripheral data signal must be configured to use pins close to the clock pin. This ensures that the
- internal paths from the peripheral to the pin have the same delay, so that the data and clock signals reach
- the pins at the same time.
- For high-speed signals, the printed circuit board (PCB) layout must use short PCB traces of identical length. This makes sure any delays are kept to a minimum and it assures close to identical delay and clock path.

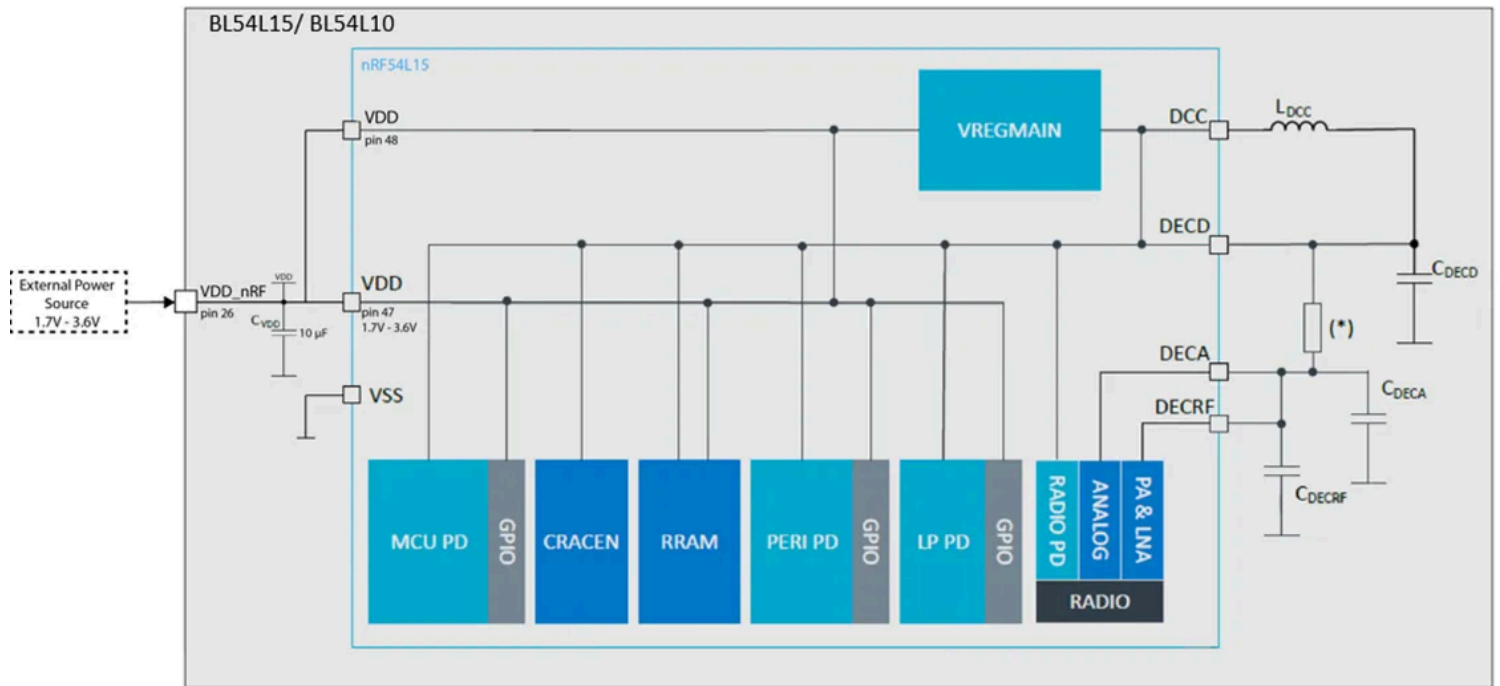
Note 3 (Dedicated pins):

- **UARTE20/21:** Can use any pin on GPIO port P1. Can be connected across power domains to dedicated pin on P2.
- **SPIM00:** Has dedicated pins on GPIO port P2. For 32MHz operation, the pins must be configured using extra high drive E0/E1 configuration.
- **SPIM20/21:** Can use any pin on GPIO port P1. Can be connected across power domains to dedicated pin on P2.
- **SPIS20/21:** Can use any pin on GPIO port P1. Can be connected across power domains to dedicated pin on P2.
- **TRACE:** Has dedicated pins on GPIO port P2. For 32MHz operation, the pins must be configured using extra high drive E0/E1 configuration.
- **GRTC:** Has dedicated pins for clock and PWM output.
- **TAMPC:** Has dedicated pins for active shield inputs and outputs.
- **FLPR:** Uses dedicated pins on GPIO port P2 for emulated peripherals such as QSPI.
- **RADIO:** Uses dedicated pins on GPIO port P1 for antenna switch control (DFEGPIO for direction finding).
- **NFC:** Uses dedicated pins listed in pin definitions table.

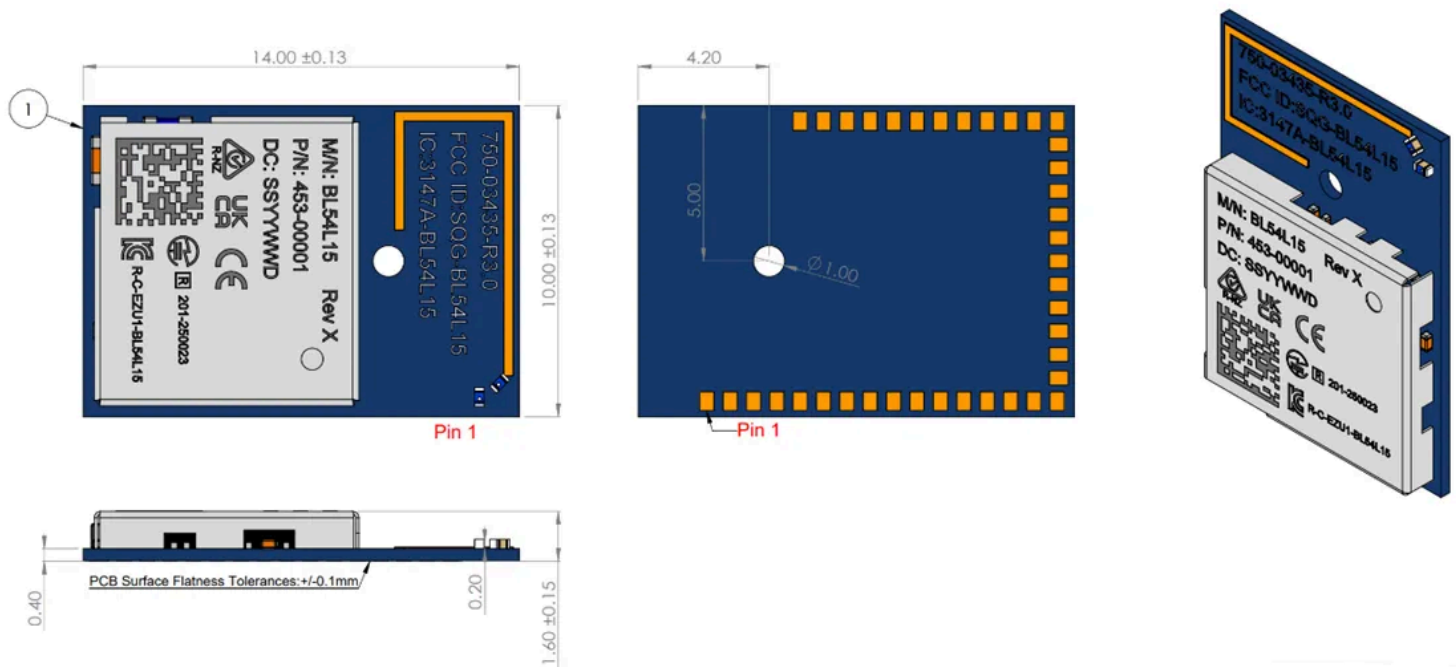
Note 4 (SWDIO / SWCLK / nRESET / VDD / GND):

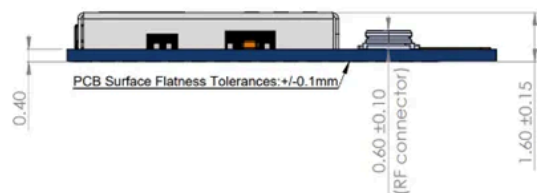
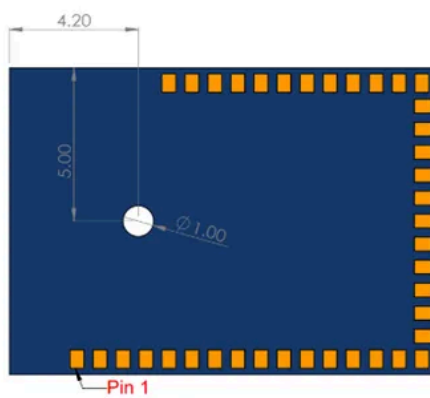
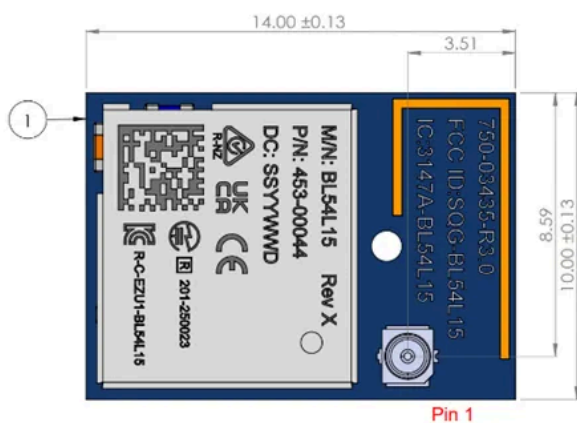
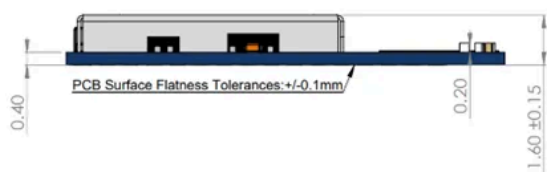
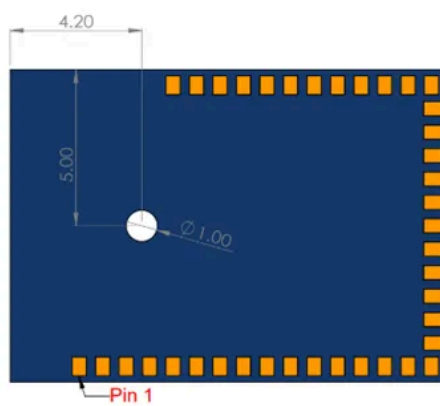
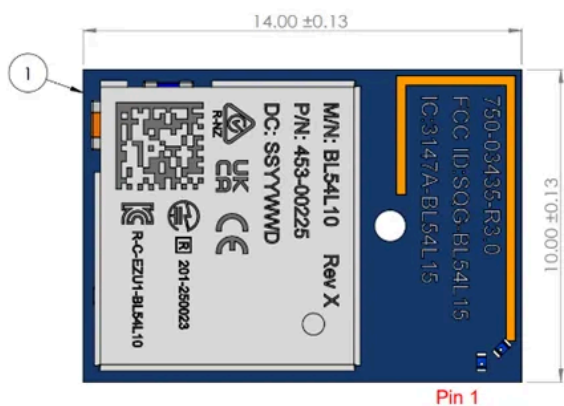
- Customer MUST bring out SWDIO, SWCLK, nRESET, VDD, GND for programming purposes.

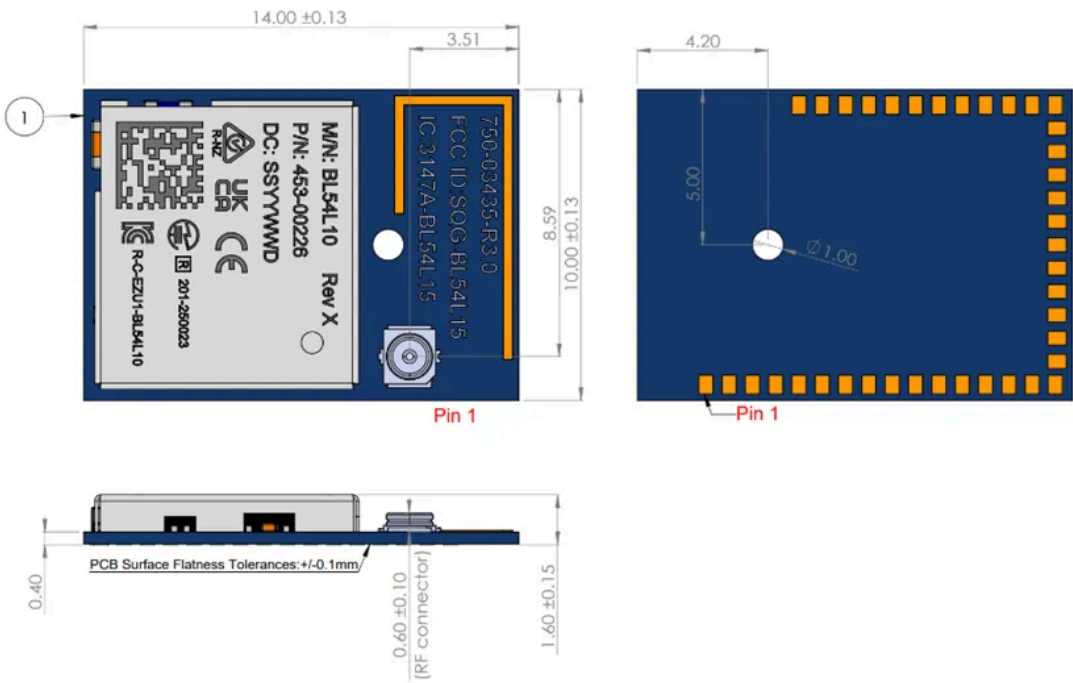
Power Supply



Mechanical Drawings







3D models for BL54L15 Module, MHF4 (453-00044) and BL54L15 Module, Trace Antenna (453-00001) on the BL54L15 product page – <https://www.ezurio.com/product/bl54l15-series-bluetooth-le-80215-4-nfc>

Electrical Characteristics

Absolute Maximum Ratings

Absolute maximum ratings are the extreme limits for supply voltage and voltages on digital and analogue pins of the module are listed below; exceeding these values causes permanent damage.

Parameter	Min	Max	Unit
Supply Voltages			
VDD_nRF	-0.3	3.9	V
VDD_nRF_ext	-0.3	3.7	V
I/O pin voltage			
Voltage at GPIO pin (at VDD≤3.6V)	-0.3	VDD_nRF + 0.3	V
Voltage at GPIO pin (at VDD>3.6V)		3.9	V
Voltage at GPIO pin under extended operating temperature (at VDD≤3.4V)	-0.3	VDD_nRF + 0.3	V

Voltage at GPIO pin under extended operating temperature (at VDD > 3.4V)	-0.3	3.7	V
NFC antenna pin current (NFC1/2)	-	130	mA
Environmental			
Storage temperature	-40	+125	°C
MSL (Moisture Sensitivity Level)	-	4	-
ESD (as per EN301-489)		4	kV
Conductive		8	kV
Air Coupling			
Flash Memory (Endurance) (Note 2)	10,000		Write/erase cycles
Flash Memory (Retention)	10 years at 85°C 2 years at 105°C		years at TBD °C

Note 1: The absolute maximum rating for VDD pin (max) is 3.9 V for the BL54L15.
Note 2: Wear levelling can be implemented by customer.

Recommended Operating Conditions

Power supply operating parameters

Parameter	Min	Typ	Max	Unit
VDD_nRF (independent of DCDC) supply range	1.7		3.6	V
VDD_nRF_ext (VDD_nRF supply voltage under extended operating temperature)	1.7		3.4	V
VDD_nRF Maximum ripple or noise (See Note 1)	-	-	10	mV
Operating Temperature Range	-40	+25	+85	°C
Extended operating temperature	85		105	°C

Note 1: This is the maximum VDD_nRF ripple or noise (at any frequency) that does not disturb the radio.

Note 2: The on-board power-on reset circuitry may not function properly for rise times longer than the specified maximum.

Note 3: Some electrical parameters are valid only for the operating temperature range conditions. When this is the case, an additional parameter for the extended operating temperature condition is provided.

Power Management & Consumption

Power Consumption

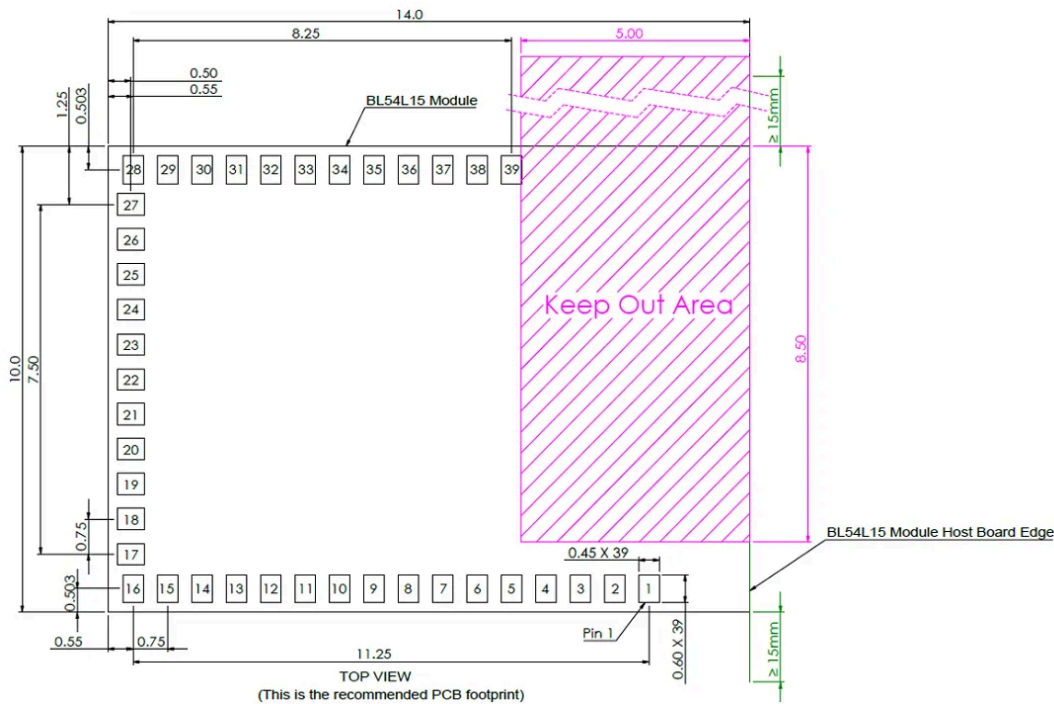
Active Modes Peak Current (for max Tx power +7dBm) – Radio only	30 mA peak Tx, 15mA avg TX@ 1.8V, LE1M	
	25 mA peak Tx, 23mA avg TX@ 1.8V, CW	
	17 mA peak Tx, 8.3mA avg TX@ 3.3V, LE1M	
	14 mA peak Tx, 12.6mA avg TX@ 3.3V, CW	
Active Modes Peak Current (for Tx power -40dBm) – Radio only	8 mA peak Tx, 2.5mA avg TX @ 1.8V, LE1M	
	7 mA peak Tx, 3.1mA avg TX@ 1.8V, CW	
	6.5 mA peak Tx, 2.1mA avg TX @ 3.3V, LE1M	
	5 mA peak Tx, 2.6mA avg TX @ 3.3V, CW	
Active Modes Average Current	Depends on many factors.	
Ultra-low Power Modes	System ON Idle	3 uA (System ON IDLE with GRTC (XOSC) and 256 KB RAM) 2.6uA (System ON IDLE with GRTC (XOSC) and 192 KB RAM)
	System OFF	0.6 uA

Integration Guidelines

Host PCB Land Pattern and Antenna Keep-Out

PCB footprint - BL54L15 (DXF and Altium format) and SCH Symbol - BL54L15 (Altium format) can be found on the BL54L15 product page – <https://www.ezurio.com/product/bl54l15-series-bluetooth-le-80215-4-nfc>

BL54L10 use the same footprint of BL54L15.



All dimensions are in mm.

Notes:

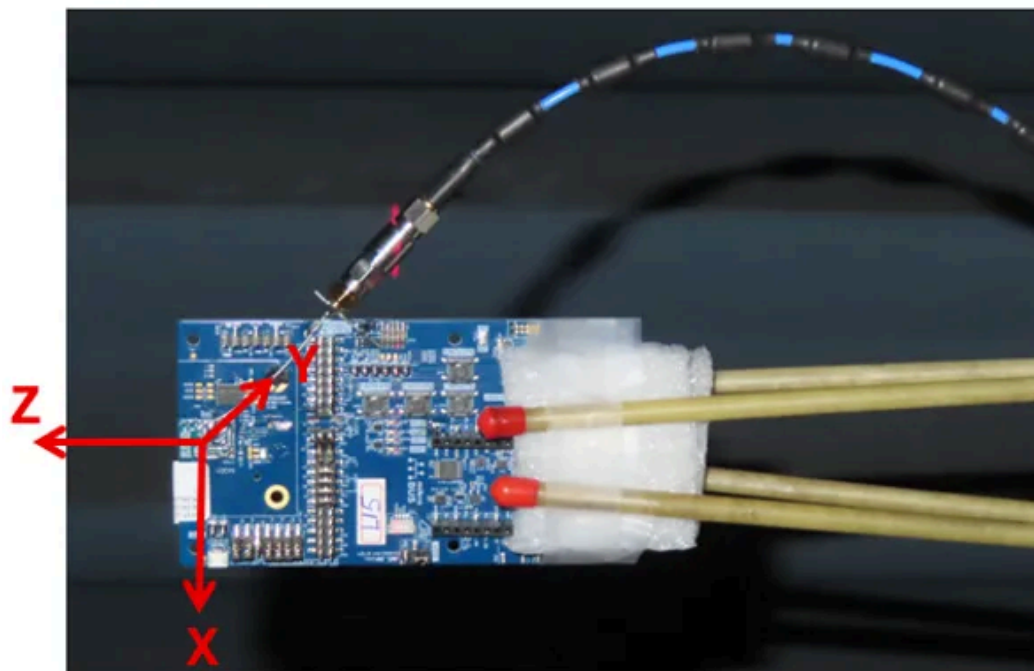
1. Ensure there is no copper in the antenna 'keep out area' on any layers of the host PCB. Also keep all mounting hardware or any metal clear of the area (Refer to 7.3.2) to reduce effects of proximity detuning the antenna and to help antenna radiate properly.
2. For the best on-board antenna performance, the module 453-00001/453-00225 MUST be placed on the edge of the host PCB and preferably in the edge center and host PCB, the antenna "Keep Out Area" is extended (see Note 4).
3. BL54L15 development board has the 453-00001 placed on the edge of the PCB board (and not in corner) for that the Antenna keep out area is extended down to the corner of the development board, see section 8.2 Host PCB Land Pattern and Antenna Keep-out for the 453-00001 and 453-00225.
4. Ensure that there is no exposed copper under the module on the host PCB.
5. You may modify the PCB land pattern dimensions based on their experience and/or process capability.

On-Board PCB Trace Antenna Characteristics

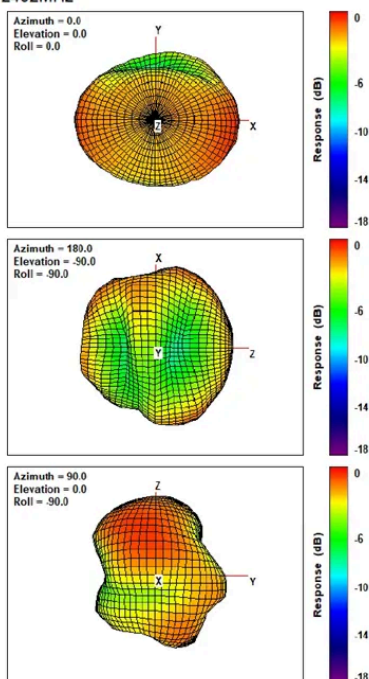
Summary of Antenna Performance

	2402 MHz Peak	2402 MHz Avg	2440 MHz Peak	2440 MHz Avg	2480 MHz Peak	2480 MHz Avg
453-00001/453-00225 PCB trace antenna	+0.2dBi	-4.24dBi	+1.49dBi	-3.19dBi	+1.39dBi	-3.27dBi

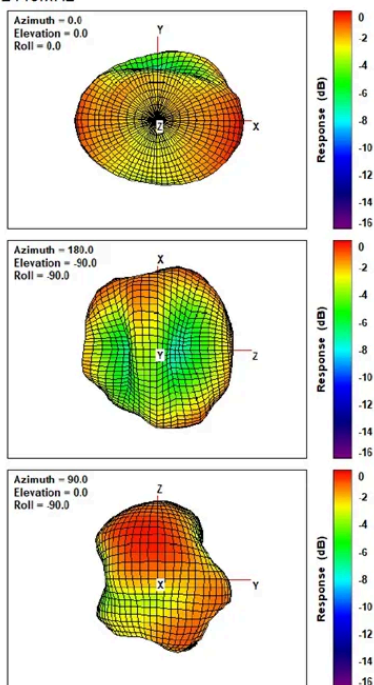
Radiated Performance



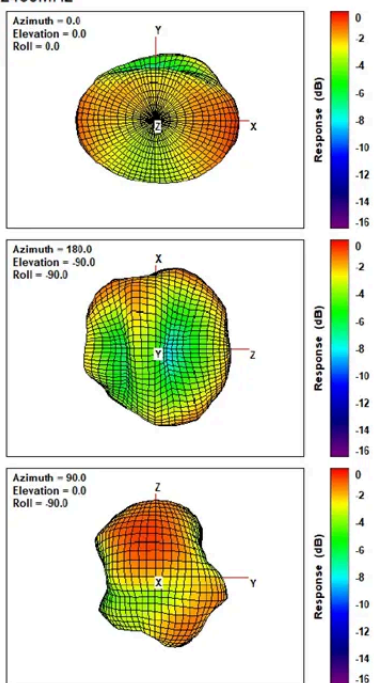
2402MHz



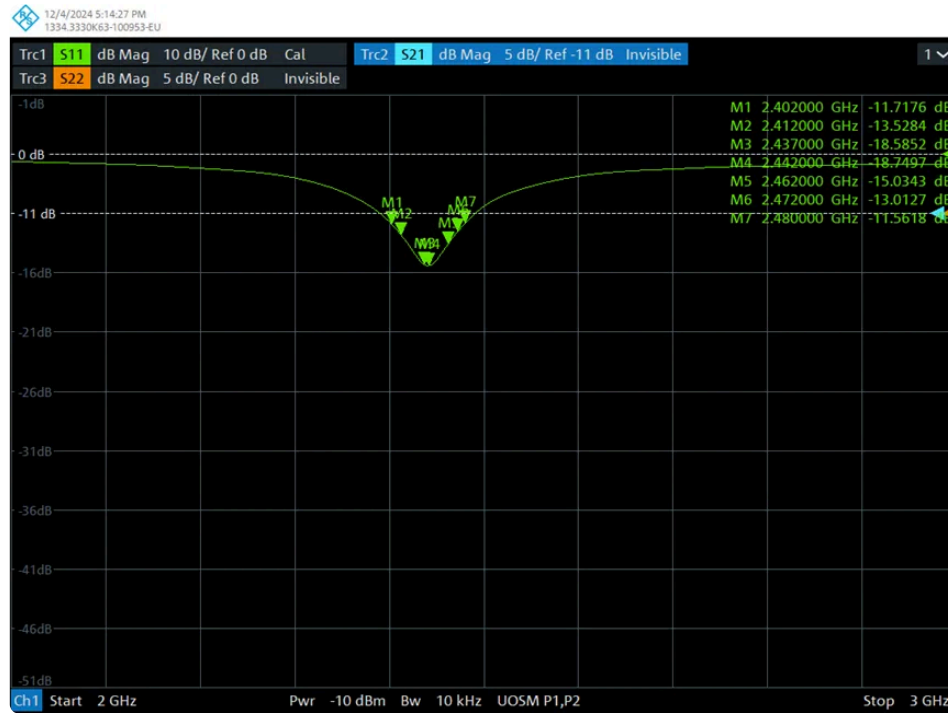
2440MHz



2480MHz



Antenna S11 Measuring Data



Software Requirements Related to Hardware

32MHz crystal internal load capacitor setting

BL54L15/ BL54L10 module contains the 32 MHz crystal but the load capacitors to create 32 MHz crystal oscillator circuit are inside the nRF54L15/ nRF54L10 chipset. Customer can set the internal nRF54L15/nRF54L10 capacitors from 4 pF to 17 pF in 0.25pF if needed, or use default value without changing.

Circuit (Overview and Checklist)

The BL54L15/ BL54L10 is easy to integrate, requiring one mandatory external 10uF capacitor on customers board and apart from that those components which customer require for development and in your end application.

The following are suggestions for your design for the best performance and functionality.

Checklist (for Schematic):

- BL54L15/ BL54L10 power supply:**
 Normal voltage mode power supply mode is entered when the external supply voltage (1.7V-3.6V) is connected to both VDD_nRF pin (pin26). External power source should be within the operating range, rise time and noise/ripple specification of the BL54L15/ BL54L10. Add decoupling capacitors for filtering the external source. Power-on reset circuitry within BL54L15/ BL54L10 series module incorporates brown-out detector, thus simplifying your power supply design. Upon application of power, the internal power-on reset ensures that the module starts correctly.
- AIN (ADC) and GPIO pin IO voltage levels:**
 BL54L15/BL54L10 GPIO voltage levels are at VDD_nRF. Ensure input voltage levels into GPIO pins are at VDD_nRF also. Ensure ADC pin maximum input voltage for damage is not violated.

- **AIN (ADC) impedance and external voltage divider setup:**

If you need to measure with ADC a voltage higher than 3.6V, you can connect a high impedance voltage divider to lower the voltage to the ADC input pin.

- **SWD:**

This is REQUIRED for loading firmware. MUST wire out the SWD two wire interface on host design. Five lines should be wired out, namely SWDIO, SWDCLK, nRESET, GND and VDD.

- **UART and flow control (CTS, RTS):**

Required if customer requires UART.

- **TWI (I2C):**

It is essential to remember that pull-up resistors on both SCL and SDA lines are required, the value as per I2C standard. nRF54L15/ nRF54L10 can provide 13K Ohms typical pull up values internally. For other values, fit external pull-up resistor on both SCL and SDA as per I2C specification to set speed. The I2C specification allows a line capacitance of 400pF.

- **QSPI, High Speed SPI, High speed TWI (I2C, 1Mbps) and Trace:**

High-Speed SPI, TWI and Trace come on dedicated GPIO pins only. Other lower speed SPI and TWI can come out on any GPIO pins.

For all high-speed signal, the printed circuit board (PCB) layout must ensure that connections are made using short PCB traces.

- **GPIO pins**

If GPIO is selected as an input, ensure the input is not floating (which can cause current consumption to drive with time in low power modes (such as System ON Idle), by selecting the internal pull up or pull down.

- **NFC antenna connector:** To make use of the Ezurio flexi-PCB NFC antenna (part # 0600-00061), fit connector:

Description – FFC/FPC Connector, Right Angle, SMD/90d, Dual Contact, 1.2 mm Mated Height

Manufacturer – Molex

Manufacturers Part number – 512810594

Add tuning capacitors of 300 pF on NFC1 pin to GND and 300 pF on NFC2 pins to GND if the PCB track length is similar as development board.

- **nRESET pin (active low):**

Hardware reset. Wire out to push button or drive by host.

By default module is out of reset when power applied to VDD_nRF pins (13K pull-up inside BL54L15/ BL54L10 (nRF54L15-QFAA/nRF54L10-QFAA)).

- **Optional External 32.768kHz crystal:**

If the optional external 32.768kHz crystal is needed, then use a crystal that meets specification and add load capacitors (either inside nRF54L15-QFAA/nRF54L10-QFAA or discrete capacitors outside BL54L15/BL54L10 (nRF54L15-QFAA/nRF54L10-QFAA) whose values should be tuned to meet all specification for frequency and oscillation margin.

PCB Layout

PCB Layout on Host PCB - General

Checklist (for PCB):

- MUST locate BL54L15/ BL54L10 module close to the edge of PCB (mandatory for the 453-00001/453-00225 for on-board PCB trace antenna to radiate properly).

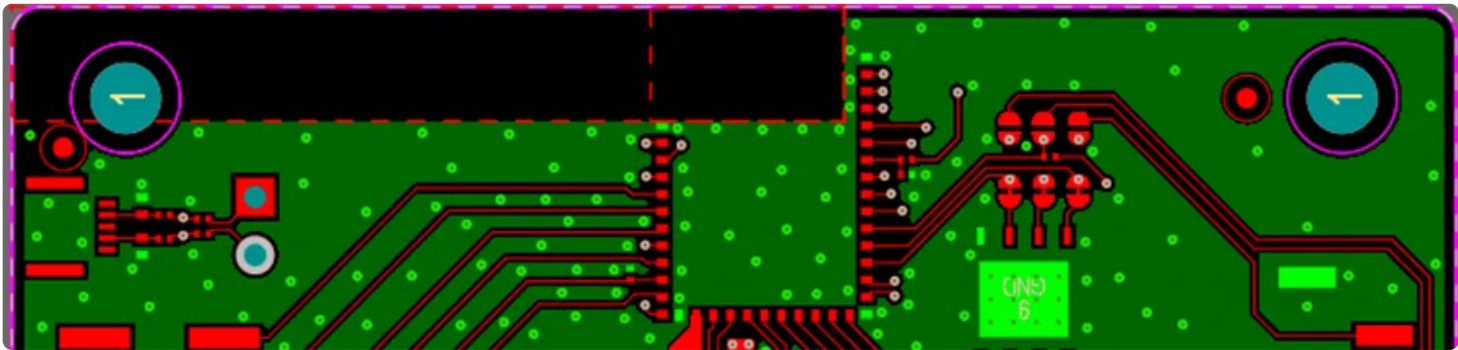
- Use solid GND plane on inner layer (for best EMC and RF performance).
- All module GND pins MUST be connected to host PCB GND.
- Place GND vias close to module GND pads as possible.
- Unused PCB area on surface layer can be flooded with copper but place GND vias regularly to connect the copper flood to the inner GND plane. If GND flood copper is on the bottom of the module, then connect it with GND vias to the inner GND plane.
- Route traces to avoid noise being picked up on VDD_nRF supply and AIN (analogue), GPIO (digital) traces and high-speed traces.
- Ensure no exposed copper is on the underside of the module (refer to land pattern of BL54L15/BL54L10 development board).

Antenna Keep-Out on Host PCB

The 453-00001/453-00225 has an integrated PCB trace antenna and its performance is sensitive to host PCB. It is critical to locate the 453-00001/453-00225 on the edge of the host PCB (or corner) to allow the antenna to radiate properly. Refer to guidelines in section PCB land pattern and antenna keep-out area for the 453-00001/453-00225. Some of those guidelines repeated below.

- Ensure there is no copper in the antenna keep-out area on any layers of the host PCB. Keep all mounting hardware and metal clear of the area to allow proper antenna radiation.
- For best antenna performance, place the 453-00001/453-00225 module on the edge of the host PCB, preferably in the edge center.
- The BL54L15 development board (453-00001-K1) has the 453-00001 module on the edge of the board (not in the corner). The antenna keep-out area is defined by the BL54L15 development board which was used for module development and antenna performance evaluation is shown in Figure 6, where the antenna keep-out area is ~5mm wide, ~28.6mm long; with PCB dielectric (no copper) height ~1.57mm sitting under the 453-00001 PCB trace antenna module. BL54L10 module (453-00225) please follow the same design of BL54L15.
- The 453-00001 PCB trace antenna is tuned when the 453-00001 is sitting on development board (host PCB) with size of 113 mm x 63.5 mm x 1.6mm.
- A different host PCB thickness dielectric will have small effect on antenna.
- The antenna-keep-out defined in the 8.2 Host PCB Land Pattern and Antenna Keep-out for the 453-00001 and 453-00225 section.

- Host PCB land pattern and antenna keep-out for the BL54L15/BL54L10 applies when the 453-00001/453-00225 is placed in the edge of the host PCB preferably in the edge center. Below shows an example.



- Note 1:

The BL54L15/BL54L10 module is placed on the edge, preferably edge centre of the host PCB.
- Note 2:

Copper cut-away on all layers in the Antenna Keep-out area under the 453-00001/453-00225 on host PCB.

Antenna Keep-out and Proximity to Metal or Plastic

Checklist (for metal /plastic enclosure):

- Minimum safe distance for metals without seriously compromising the antenna (tuning) is 40 mm top/bottom and 30 mm left or right.
- Metal close to the 453-00001/453-00225 PCB trace monopole antenna (bottom, top, left, right, any direction) will have degradation on the antenna performance. The amount of that degradation is entirely system dependent, meaning you will need to perform some testing with your host application.
- Any metal closer than 20 mm will begin to significantly degrade performance (S11, gain, radiation efficiency).
- It is best that you test the range with a mock-up (or actual prototype) of the product to assess effects of enclosure height (and materials, whether metal or plastic) and host PCB ground (GND plane size).

External Antenna Integration with BL54L15/BL54L10 MHF4 variant (453-00044/453-00226)

Please refer to the regulatory sections for FCC, ISED, CE, MIC, UKCA and RCM details of use of BL54L15 with external antennas in each regulatory region.

The BL54L15/BL54L10 family has been designed to operate with the below external antennas (with a maximum gain of 2.32 dBi). The required antenna impedance is 50 ohms. See the table below. External antennas improve radiation efficiency.

External antennas for the BL54L15/BL54L10 RF trace pin variant module (453-00044/453-00226)

Manufacturer	Model	Ezurio Part Number	Type	Connector	Peak Gain 2400-2500 MHz	Peak Gain2400-2480 MHz
Ezurio (Laird Connectivity)	NanoBlue	EBL2400A1-10MH4L	PCB Dipole	IPEX MHF4	2 dBi	-

Ezurio (Laird Connectivity)	FlexPIFA	001-0022	PIFA	IPEX MHF4	-	2 dBi
Mag.Layers	EDA-8709-2G4C1-B27-CY	0600-00057	Dipole	IPEX MHF4	2.32 dBi	-
Ezurio (Laird Connectivity)	mFlexPIFA	EFA2400A3S-10MH4L	PIFA	IPEX MHF4	-	2 dBi
Ezurio (Laird Connectivity)	iFlexPIFA Mini	EFG2401A3S-10MH4L	PIFA	IPEX MHF4	-	2 dBi
Ezurio (Laird Connectivity)	Ezurio NFC	0600-00061	NFC	N/A	-	-

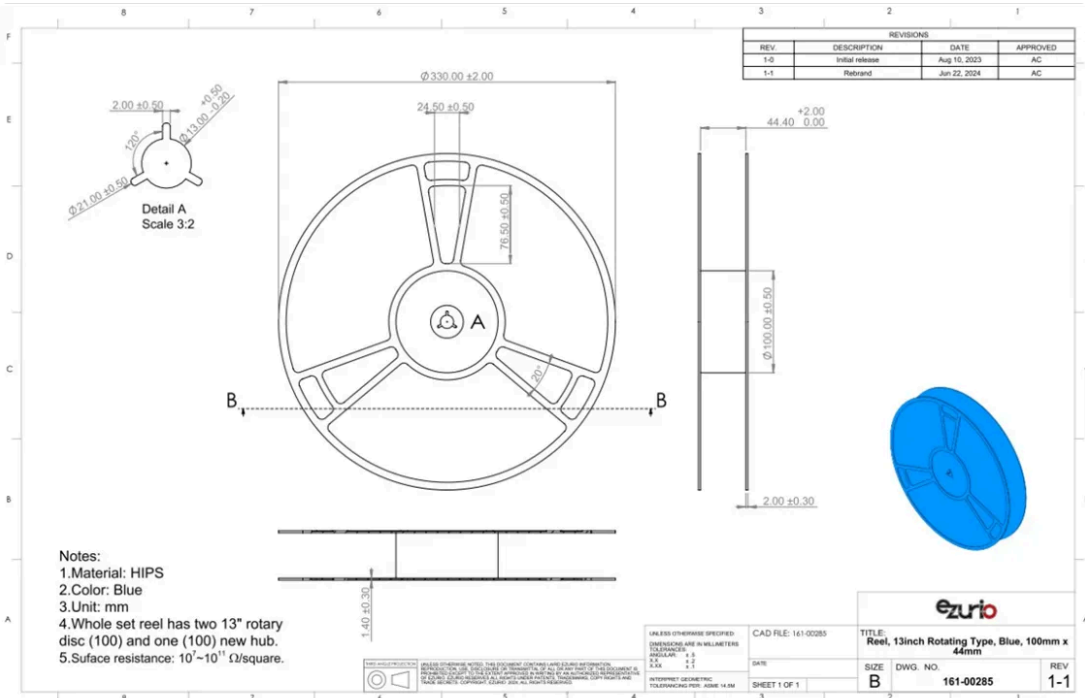
Application Note for Surface Mount Modules

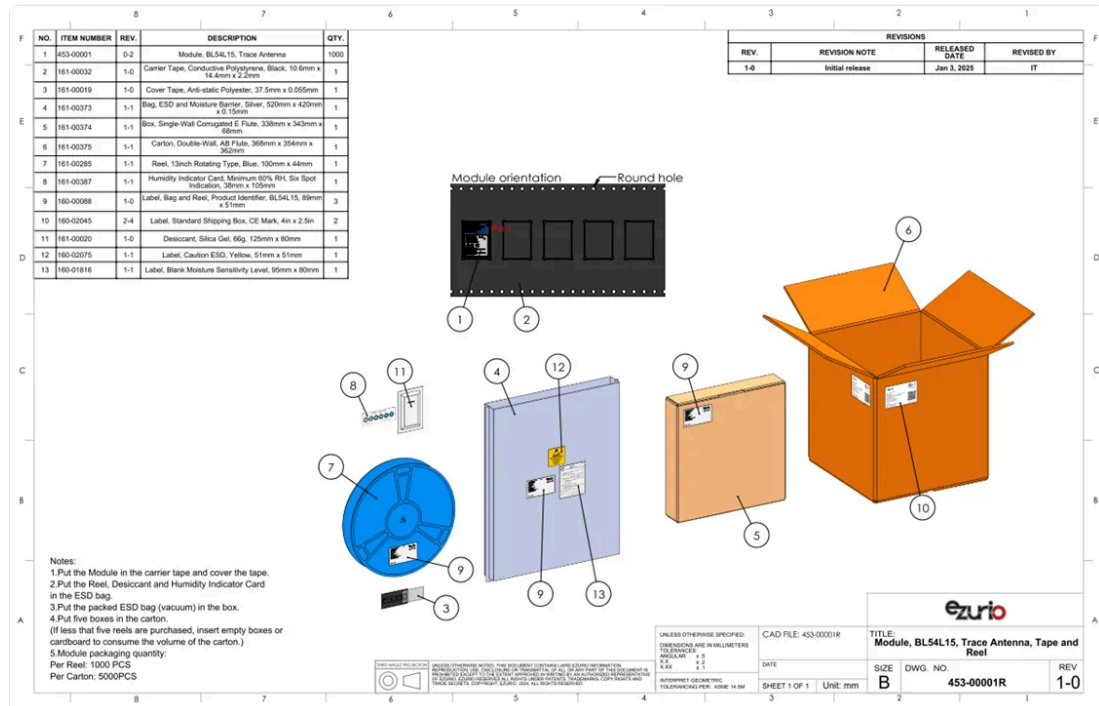
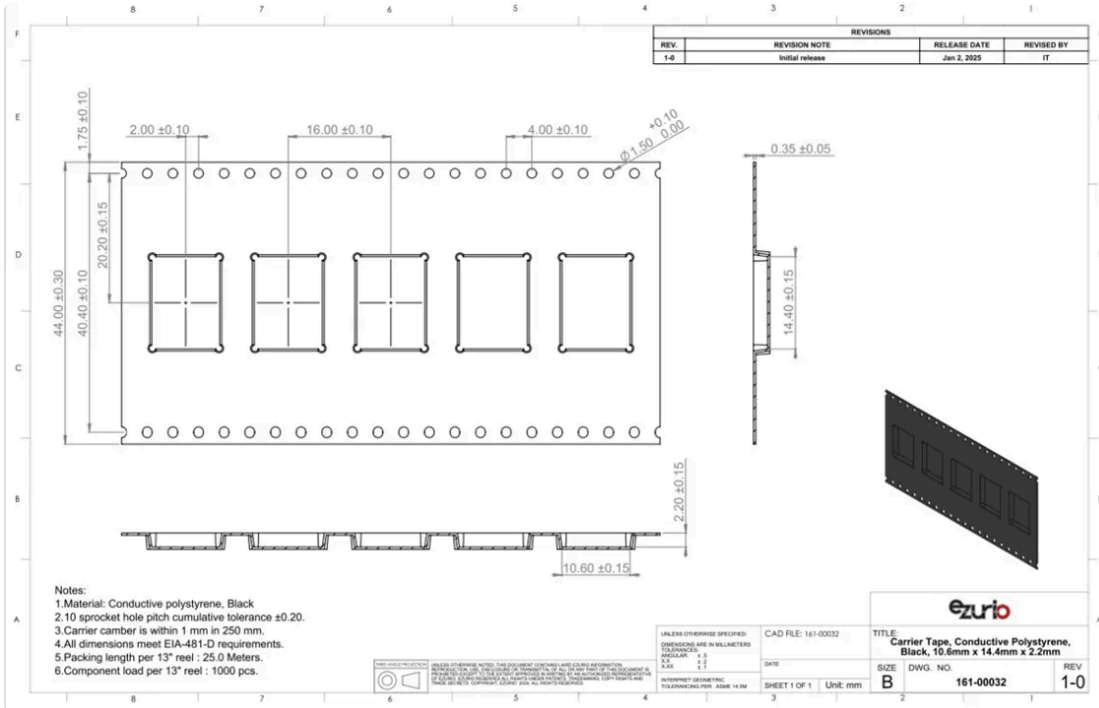
Introduction

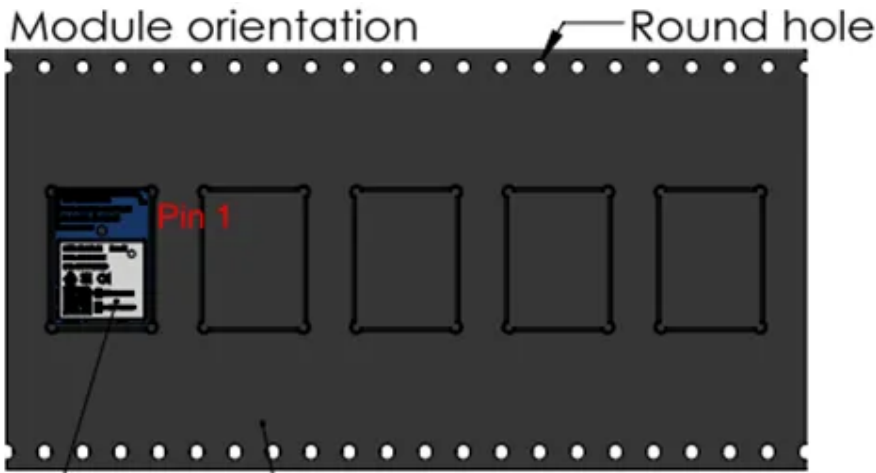
Ezurio's surface mount modules are designed to conform to all major manufacturing guidelines. This application note is intended to provide additional guidance beyond the information that is presented in the user manual. This application note is considered a living document and will be updated as new information is presented.

The modules are designed to meet the needs of several commercial and industrial applications. They are easy to manufacture and conform to current automated manufacturing processes.

Part numbers – **453-00001R/453-00225R** and **453-00044R/453-00226R** are shipped as Tape / Reel, with a reel containing 1,000 pcs.







Shipping

All modules are shipped in tape and reel package and sealed in ESD Bags.

Labeling

MSL Label

Caution
 This bag contains
MOISTURE-SENSITIVE DEVICES

LEVEL

If blank, see adjacent bar code label

- Calculated shelf life in sealed bag : 12 months at <40 °C and <90% relative humidity (RH)
- Peak package body temperature: _____ °C
If blank, see adjacent bar code label
- After bag is opened, devices that will be subjected to reflow solder or other high temperature process must be
 - Mounted within: _____ hours of factory conditions
If blank, see adjacent bar code label
 ≤30 °C/60% RH, or
 - Stored per J-STD-033
- Devices require bake, before mounting, if:
 - Humidity Indicator Card reads > 10% for level 2a - 5a devices or > 60% for level 2 devices when read at 23 ± 5 °C
 - 3a or 3b are not met.
- If baking is required, refer to IPC/JEDEC J-STD-033 for bake procedure.

Bag Seal Date: _____
If blank, see adjacent bar code label

Note: Level and body temperature defined by IPC/JEDEC J-STD-020

Product Identifier Labels

Ezurio Part NO: 453-00001R P/N: 9901142054 Quantity: XXX PCS Date Code: SSYYWWD PACKING NO: RVNYYWWXXXX	Rev 1 ezurio  201-250023  R-C-EZU1-BL54L15    Made in Vietnam
--	---

Ezurio Part NO: 453-00044R P/N: 9901142055 Quantity: XXX PCS Date Code: SSYYWWD PACKING NO: RVNYYWWXXXX	Rev 1 ezurio  201-250023  R-C-EZU1-BL54L15    Made in Vietnam
--	--

Ezurio Part NO: 453-00225R P/N: 9901150000 Quantity: XXX PCS Date Code: SSYYWWD PACKING NO: RVNYYWWXXXX	Rev X ezurio  201-250023  R-C-EZU1-BL54L10    Made in Vietnam
--	---



The following labels are placed on the pizza box.



Ezurio Part NO: 453-00225R

P/N: 9901150000

Quantity: XXX PCS

Date Code: SSYYWWD

PACKING NO: RVNYYWWXXXX

Rev X

ezurio

201-250023

R-C-EZU1-BL54L10

Made in Vietnam

Ezurio Part NO: 453-00226R

P/N: 9901150001

Quantity: XXX PCS

Date Code: SSYYWWD

PACKING NO: RVNYYWWXXXX

Rev X

ezurio

201-250023

R-C-EZU1-BL54L10

Made in Vietnam

The following labels are placed on the master shipping carton.

ezurio

P/N: 453-00001R

Revision: 1

Description: Module, BL54L15, Trace Antenna, Tape and Reel

Quantity: XXX PCS

Date Code: SSYYWWD

Country of Origin: VN

PACKING NO: RVNYYWWXXXX

Ezurio
50 S.Main St. Ste. 1100
Akron, OH 44308

201-250023

R-C-EZU1-BL54L15

PO: XXXXXX

www.ezurio.com



P/N: 453-00044R
Revision: 1
Description: Module, BL54L15, MHF4, Tape and Reel
Quantity: XXX PCS
Date Code: SSYYWWD
Country of Origin: VN
PACKING NO: RVNYYWWXXXX




R-NZ
 
UKCA
 
CE


 201-250023

 R-C-EZU1-BL54L15

PO: XXXXXX



www.ezurio.com

Ezurio
50 S.Main St. Ste. 1100
Akron, OH 44308



P/N: 453-00225R
Revision: X
Description: Module, BL54L10, Trace Antenna, Tape and Reel
Quantity: XXX PCS
Date Code: SSYYWWD
Country of Origin: VN
PACKING NO: RVNYYWWXXXX




R-NZ
 
UKCA
 
CE


 201-250023

 R-C-EZU1-BL54L10

PO: XXXXXX



www.ezurio.com

Ezurio
50 S.Main St. Ste. 1100
Akron, OH 44308



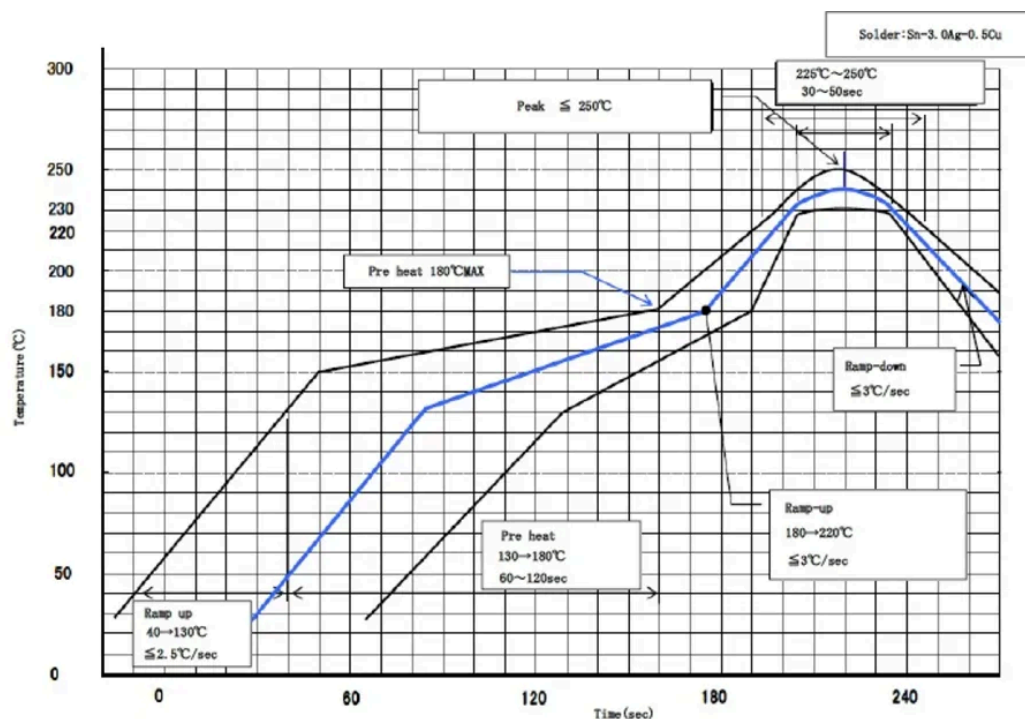
Recommended Stencil Aperture

When soldering, the stencil thickness should be ≥ 0.1 mm.

Reflow Parameters/ Soldering

Convection reflow or IR/Convection reflow (one-time soldering or two-time soldering in air or nitrogen environment)

- Measuring point – IC package surface
- Temperature profile:



- Ramp-up: $40 \sim 130^{\circ}\text{C}$. Less than 2.5°C/sec

- Pre heat: 130-180 °C 60-120 sec, 180 °C MAX
- Ramp-up: 180-220 °C. Less than 3 °C/sec
- Peak Temperature: MAX 250 °C
 - 225 °C ~ 250 °C, 30 ~ 50 sec
- Ramp-down: Less than 3 °C/sec

Cautions when Removing the BL54L15/BL54L10 from the Platform for RMA

- Bake the platform PCBA before removing the BL54L15/BL54L10 module from the platform.
- Remove the BL54L15/BL54L10 module by using a hot air gun. This process should be carried out by a skilled technician.

Recommended conditions for one-side component platform:

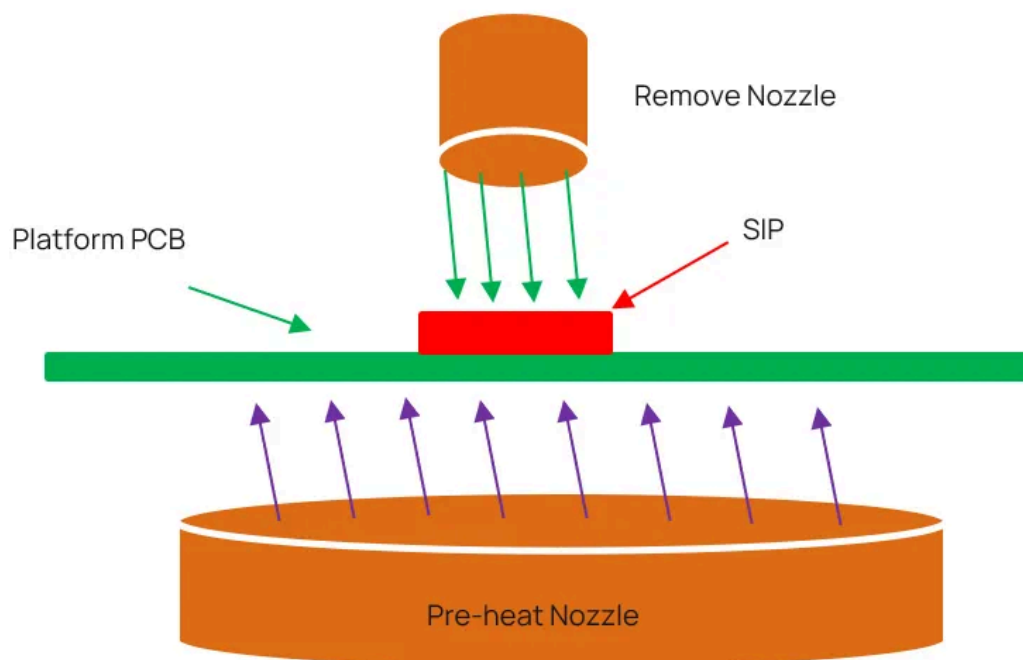
- Set the hot plate at 280°C.
- Put the platform on the hot plate for 8~10 seconds.
- Remove the device from platform.



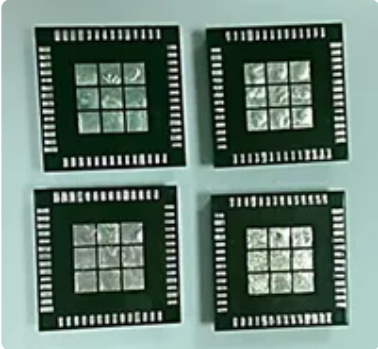
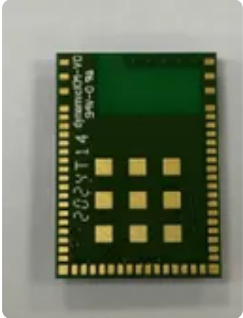
Recommended conditions for two-side components platform:

- Use two hot air guns.
- On the bottom, use a pre-heated nozzle (temp setting of 200~250°C) at a suitable distance from the platform PCB.

- On the top, apply a remove nozzle (temp setting of 330°C). Heat until device can be removed from platform PCB.



- Remove the residue solder under the bottom side of device. (Note: Alternate module pictured as an example)

	
(Not accepted for RMA)	(Accepted for RMA analysis)

- Remove and clean the residue flux as needed.

Precautions for Use

- Opening/handing/removing must be done on an anti-ESD treated workbench. All workers must also have undergone anti-ESD treatment.
- The devices should be mounted within one year of the date of delivery.
- The BL54L15/BL54L10 modules are MSL 4 rated.

Environmental and Reliability

Environmental Requirements

Required Storage Conditions

Prior to Opening the Dry Packing

The following are required storage conditions *prior to opening the dry packing*:

- Normal temperature: 5~40 °C
- Normal humidity: 80% (Relative humidity) or less
- Storage period: One year or less

Note: Humidity means relative humidity.

After Opening the Dry Packing

The following are required storage conditions *after opening the dry packing* (to prevent moisture absorption):

- Storage conditions for one-time soldering:
 - Temperature: 5-25°C
 - Humidity: 60% or less
 - Period: 72 hours or less after opening
- Storage conditions for two-time soldering
 - Storage conditions following opening and prior to performing the 1st reflow:
 - Temperature: 5-25°C
 - Humidity: 60% or less
 - Period: A hours or less after opening
 - Storage conditions following completion of the 1st reflow and prior to performing the 2nd reflow
 - Temperature: 5-25°C
 - Humidity: 60% or less
 - Period: B hours or less after completion of the 1st reflow

Note: Should keep A+B within 72 hours.

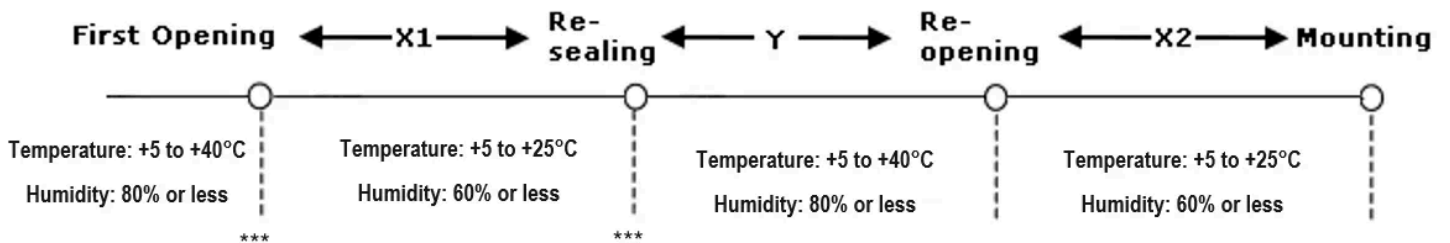
Temporary Storage Requirements after Opening

The following are temporary storage requirements after opening:

- Only re-store the devices once prior to soldering.
- Use a dry box or place desiccant (with a blue humidity indicator) with the devices and perform dry packing again using vacuumed heat-sealing.

The following indicate the required storage period, temperature, and humidity for this temporary storage:

- Storage temperature and humidity:



- Storage period:
 - X1+X2 – Refer to Material handling information
 - Required Storage Conditions. Keep is X1+X2 within 72 hours.
 - Y – Keep within two weeks or less.

Baking Conditions

Baking conditions and processes for the module follow the J-STD-033 standard which includes the following:

- The calculated shelf life in a sealed bag is 12 months at < 40°C and < 80% relative humidity.
- Once the packaging is opened, the SiP must be mounted (per **MSL4/Moisture Sensitivity Level 4**) within 72 hours at < 30 °C and < 60% relative humidity.

If the SiP is not mounted within 72 hours or if, when the dry pack is opened, the humidity indicator card displays >10% humidity, then the product must be baked for 48 hours at 125 °C (±5 °C).

Reflow Profile

Reliability Tests

Climatic and Dynamic

Climatic and Dynamic Reliability Test Results for BL54L15/BL54L10 Modules

Test Item	Specification	Standard	Test Result
Thermal Shock	Temperature: -40 ~ 105°C Ramp time: Less than 10 seconds. Dwell Time: 10 minutes Number of Cycles: 350 times	*JESD22-A106 *IEC 60068-2-14 for dwell time and number of cycles	PASS
Vibration Non-Operating Unpackaged device	Vibration Wave Form: Sine Waveform Vibration frequency / Displacement: 20-80 Hz/1.5mm Vibration frequency / Acceleration: 80-2000 Hz/20g Cycle Time: 4 min/cycle Number of Cycles: 4 cycle/axis Vibration Axes: X, Y and Z (Rotate each axis on vertical vibration table)	JEDEC 22-B103B (2016)	PASS

Mechanical Shock	Pulse shape: Half-sine waveform	JEDEC 22-B110B.01 (2019)	PASS
Non-Operating Unpackaged device	Impact acceleration: 1500 g		
	Pulse duration: 0.5 ms		
	Number of shocks: 30 shocks (5 shocks for each face)		
	Orientation: Bottom, top, left, right, front, and rear faces		

Reliability Prediction

MTBF Predictions for BL54L15/BL54L10 Modules

Ezurio Part Number	Environment	Standard	Test Result 45 °C (Hours)
453-00001R 453-00001C 453-00225R 453-00225C	Ground, Fixed, Uncontrolled	Telcordia Issue 4	1,559,211
453-00044R 453-00044C 453-00226R 453-00226C	Ground, Fixed, Uncontrolled	Telcordia Issue 4	1,559,211
453-00001R 453-00001C 453-00225R 453-00225C	Mobile, Fixed, Uncontrolled	Telcordia Issue 4	605,957

453-00044R	Mobile, Fixed, Uncontrolled	Telcordia Issue 4	605,957
453-00044C			
453-00226R			
453-00226C			
Ezurio Part Number	Environment	Standard	Test Result 105 °C (Hours)
453-00001R	Ground, Fixed, Uncontrolled	Telcordia Issue 4	980,517
453-00001C			
453-00225R			
453-00225C			
453-00044R	Ground, Fixed, Uncontrolled	Telcordia Issue 4	980,517
453-00044C			
453-00226R			
453-00226C			
453-00001R	Mobile, Fixed, Uncontrolled	Telcordia Issue 4	283,138
453-00001C			
453-00225R			
453-00225C			
453-00044R	Mobile, Fixed, Uncontrolled	Telcordia Issue 4	283,138
453-00044C			
453-00226R			
453-00226C			

Regulatory, Qualification & Certifications

Regulatory Approvals

Full regulatory information on the BL54L15/BL54L10, including the Regulatory Information Guide, grants, and test reports are available on the [BL54L15 product page](#) and [BL54L10 product page](#).

The BL54L15/BL54L10 holds current certifications in the following countries:

BL54L15 Certifications

Country/Region	Regulatory ID
USA (FCC)	SQG-BL54L15
EU (ETSI)	N/A (No ID Number Required)
UKCA	N/A (No ID Number Required)
Canada (ISED)	3147A-BL54L15
Japan (MIC)	201-250023
Australia (RCM)	N/A
New Zealand (RCM)	N/A
Korea	R-C-EZU1-BL54L15 (for BL54L15)
	R-C-EZU1-BL54L10 (for BL54L10)

Certified Antennas

Model	MPN	Manufacturer	Type	Connector	Peak Gain (2400 – 2500 MHz)
NanoBlue	EBL2400A1-10MH4L	Ezurio	PCB Dipole	IPEX MHF4	2 dBi
FlexPIFA	001-0022	Ezurio	FlexPIFA	IPEX MHF4L	2 dBi
EDA-8709-2G4C1-B27-CY	EDA-8709-2G4C1-B27-CY (Ezurio Part #0600-00057)	MAG.LAYERS	Dipole	IPEX MHF4	2.32 dBi
mFlexPIFA	EFA2400A3S-10MH4L	Ezurio	PIFA	IPEX MHF4L	2 dBi
I-FlexPIFA Mini Series	EFG2401A3S-10MHF4L	Ezurio	I-FlexPIFA	IPEX MHF4L	2 dBi

Ezurio NFC	0600-00061	Ezurio	Coiled Inductor	FFC/FPC Connector	-
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Bluetooth SIG Qualification

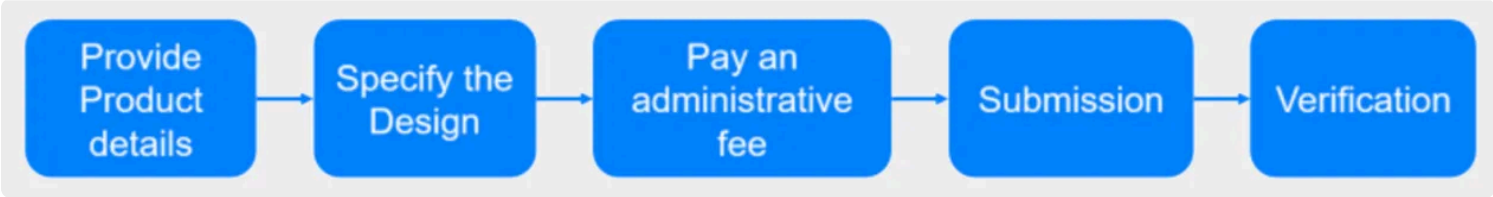
The Bluetooth Qualification Process promotes global product interoperability and reinforces the strength of the Bluetooth® brand and ecosystem to the benefit of all Bluetooth SIG members. The Bluetooth Qualification Process helps member companies ensure their products that incorporate Bluetooth technology comply with the Bluetooth Patent & Copyright License Agreement and the Bluetooth Trademark License Agreement (collectively, the Bluetooth License Agreement) and Bluetooth Specifications.

The Bluetooth Qualification Process is defined by the [Qualification Program Reference Document \(QPRD\)](#).

To demonstrate that a product complies with the Bluetooth Specification(s), each member must for each of its products:

- Identify the product, the design included in the product, the Bluetooth Specifications that the design implements, and the features of each implemented specification
- Complete the Bluetooth Qualification Process by submitting the required documentation for the product under a user account belonging to your company

The Bluetooth Qualification Process consists of the phases shown below:



To complete the Qualification Process the company developing a Bluetooth End Product shall be a member of the Bluetooth SIG. To start the application please use the following link if you are not already a Bluetooth SIG Member: [Apply for Adopter Membership](#). (Associate Membership also available which has a yearly membership fee).

Scope

This guide is intended to provide guidance on the Bluetooth Qualification Process for End Products that reference a single existing design, that has not been modified, (refer to Section 3.2.1 of the [Qualification Program Reference Document \(QPRD\)](#)).

This option applies to a Member qualifying a Product that includes an existing Design that has a DN, QDID, or DID and that Design has not been modified (e.g., rebranding a Qualified Product from another Member). The Design identified by the DN, QDID, or DID may only implement Bluetooth Specifications that are active or deprecated at the time of Submission. No modifications may be made to the Design, including changes to the ICS Form.

Changes to the Product outside the Design are allowed, including:

- Enabling technologies
- Changes to the industrial design
- Changes to the communication technology other than Bluetooth
- Changes to the features that are unrelated to Bluetooth, branding, packaging, colour, shape, Product name, or Model number

Members are responsible for assessing that modifications to the Product outside of the Design do not affect compliance with Bluetooth Specifications or result in a change to the ICS Form.

For the purposes of this document, it is assumed that the member is combining a single unmodified Core-Complete Configuration.

Qualification Steps When Referencing a single existing design, (unmodified) – Option 1 in the QPRD

For this qualification option, follow these steps:

1. To start a listing, go to: <https://qualification.bluetooth.com/>
2. Select **Start the Bluetooth Qualification Process**.
3. Product Details to be entered:
 1. Project Name (this can be the product name or the Bluetooth Design name).
 2. Product Description
 3. Model Number
 4. Product Publication Date (the product publication date may not be later than 90 days after submission)
 5. Product Website (optional)
 6. Internal Visibility (this will define if the product will be visible to other users prior to publication)
 7. If you have multiple End Products to list then you can select 'Import Multiple Products', firstly downloading and completing the template, then by 'Upload Product List'. This will populate Qualification Workspace with all your products.
4. Specify the Design:
 1. Do you include any existing Design(s) in your Product? Answer Yes, I do.
 2. Enter the single DN or QDID used in your, (for Option 1 only one DN or QDID can be referenced)
 3. Once the DN or QDID is selected it will appear on the left-hand side, indicating the layers covered by the design.
 4. Select 'I'm finished entering DN's
 5. What do you want to do next? Answer, 'Use this Design without Modification'
 6. Save and go to Product Qualification Fee
5. Test Plan and Documentation
 1. As no modifications have been made to the combined designs the tool should report the following message:
'No test plan has been generated for your new Design. Test declarations and test reports do not need to be submitted. You can continue to the next step.'
 2. Save and go to Product Qualification fee
6. Product Qualification Fee:
 1. Do you include any existing Design(s) in your Product? Answer Yes, I do.
 2. Enter the single DN or QDID used in your, (for Option 1 only one DN or QDID can be referenced)
 3. Once the DN or QDID is selected it will appear on the left-hand side, indicating the layers covered by the design.
 4. Select 'I'm finished entering DN's
 5. What do you want to do next? Answer, 'Use this Design without Modification'
 6. Save and go to Product Qualification Fee
7. Submission:
 1. Some automatic checks occur to ensure all submission requirements are complete.
 2. To complete the listing any errors must be corrected
 3. Once you have confirmed all design information is correct, tick all of the three check boxes and add your name to the signature page.

4. Now select 'Complete the Submission'.
5. You will be asked a final time to confirm you want to proceed with the submission, select 'Complete the Submission'.
6. Qualification Workspace will confirm the submission has been submitted. The Bluetooth SIG will email confirmation once the submission has been accepted, (normally this takes 1 working day).
8. Download Product and Design Details (SDoC):

1. You can now download a copy of the confirmed listing from the design listing page and save a copy in your Compliance Folder

For further information, please refer to the following webpage:

<https://www.bluetooth.com/develop-with-bluetooth/qualification-listing/>

Example Design Combinations

The following gives an example of a design possible under option 1:

Ezurio End Product design using Nordic Component based design

Design Name	Owner	Design ID	Link to listing on the SIG website	Core Spec Version
BL54L15/ BL54L15μ	Ezurio	Q345095	https://qualification.bluetooth.com/ListingDetails/274104	6.0
BL54L15/ BL54L15μ	Ezurio	Q387119	https://qualification.bluetooth.com/ListingDetails/334158	6.2

Qualify More Products

If you develop further products based on the same design in the future, it is possible to add them free of charge. The new product must not modify the existing design i.e add ICS functionality, otherwise a new design listing will be required.

To add more products to your design, select 'Manage Submitted Products' in the [Getting Started](#) page, Actions, Qualify More Products. The tool will take you through the updating process.

Ordering Information

Part Number	Product Description
453-00001R	Module, BL54L15, (Nordic nRF54L15), Trace antenna, Tape/Reel
453-00044R	Module, BL54L15, (Nordic nRF54L15), MHF4 Connector, Tape/Reel
453-00001C	Module, BL54L15, (Nordic nRF54L15), Trace antenna, Cut Tape
453-00044C	Module, BL54L15, (Nordic nRF54L15), MHF4 Connector, Cut Tape
453-00001-K1	Development kit, Module, BL54L15 (Nordic nRF54L15), Trace antenna

453-00044-K1	Development kit, Module, BL54L15 (Nordic nRF54L15), MHF4 Connector
453-00225R	Module, BL54L10, (Nordic nRF54L10), Trace antenna, Tape/Reel
453-00226R	Module, BL54L10, (Nordic nRF54L10), MHF4 Connector, Tape/Reel
453-00225C	Module, BL54L10, (Nordic nRF54L10), Trace antenna, Cut Tape
453-00226C	Module, BL54L10, (Nordic nRF54L10), MHF4 Connector, Cut Tape

Legacy - Revision History

Version	Date	Notes	Contributor(s)	Approver
0.1	10 June 2024	Initial PRELIMINARY release.	Raj Khatri	Jonathan Kaye
0.2	2 Oct 2024	Updated pin list names and notes below table1. Added section External Antenna Integration with BL54L15/BL54L10 MHF4 variant (453-00044/453-00226) Added section on Programmability.	Raj Khatri	Jonathan Kaye
0.3	24 Oct 2024	Removed Medium voltage mode (VDDM_nRF) from Specification Summary table and updated Normal voltage mode (VDD_nRF) operating range from 1.7V-2.6V to 1.7V-3.6V. Updated Block Diagram and Pin-out for 39 connection pins. Updated SCH symbol to remove pin40. Updated pin list in Table 1 (removed pin 40 VDDM_nRF). Updated BL54L15/ BL54L10 Power Supply Updated Table 4 external antenna Mag Layers EDA-8709-2G4C1-B27-CY antenna gain from 2dBi to 2.32dBi.	Raj Khatri	Jonathan Kaye

0.4	6 Dec 2024	Updated maximum Tx power to +7 dBm.	Dave Drogowski	Jonathan Kaye
0.5	11 Dec 2024	Updated sensitivity value Added trace antenna performance	Louis Chang	Jonathan Kaye
0.6	16 Dec 2024	Updated maximum Tx power to +6 dBm.	Louis Chang	Jonathan Kaye
1.0	16 Dec 2024	Initial release	Dave Drogowski	Jonathan Kaye
1.1	19 Dec 2024	Updated maximum Tx power to +7 dBm.	Louis Chang	Jonathan Kaye
1.2	13 Jan 2025	Updated title of 7.4 Updated some typo	Louis Chang	Jonathan Kaye
1.3	21 Feb. 2025	Change (VDD_nRF) operating range from 1.7V-3.6V to 1.7V-3.5V. Add VDD_nRFsupply voltage (1.7V-3.3V) under extended operating Temperature (85 to 105C) Update Absolute maximum ratings of VDD_nRF (-0.3V-3.6V)	Louis Chang	Jonathan Kaye
1.4	25 Feb. 2025	Update Minimum Transmit Power Setting to -40dBm Update Active Modes Peak Current	Louis Chang	Jonathan Kaye
1.5	27 Feb. 2025	Update 32MHz crystal oscillator internal capacitor setting value Update 32.768KHz crystal oscillator internal capacitor setting value Update NFC antenna pin current value	Louis Chang	Jonathan Kaye

1.6	3 Mar. 2025	Update Idle mode and off mode current BL54L10 Antenna performance shares with BL54L15	Louis Chang	Jonathan Kaye
1.7	10 Mar. 2025	Update Radiated Performance Update flash specs in Absolute Maximum Ratings	Louis Chang	Jonathan Kaye
1.8	18 Mar 2025	Update power consumption in 2.1	Louis Chang	Jonathan Kaye
1.9	09 Jun 2025	Update Thermal shock Max. Temperature from 85°C to 105°C of Reliability Test	Olivia Chang	Jonathan Kaye
2.0	27 Nov 2025	Update average power consumption in 2.1	Louis Chang	Jonathan Kaye
2.1	3 Dec 2025	Update Operation conditions and Absolute maximum ratings in 3.3	Louis Chang	Jonathan Kaye
2.2	16 Dec 2025	Update On -Board PCB Trace Antenna Characteristics	Louis Chang	Jonathan Kaye

Additional Information

Please contact your local sales representative or our support team for further assistance:

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Website	http://www.ezurio.com
Technical Support	http://www.ezurio.com/support
Sales Contact	http://www.ezurio.com/contact

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